

Network-Based Path Optimization for High-Frequency Cryptocurrency Arbitrage: Empirical Evidence

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Abstract. This paper presents a practical approach to reducing transaction costs in within-exchange arbitrage by trading through single-leverage contracts. Drawing on the arbitrage logic of Modigliani and Miller (1958), we formalize trading routes as alternative “capital structures” and prove the equilibrium condition $V_1 = V_2$. We further generalize this condition to $k + 1$ trading pairs and develop a network-based path-optimization framework that selects paths under conservative all-taker fees and slippage constraints using market structure rather than price prediction. Using high-frequency Binance data from January to December 2025 across 64 trading-pair groups, we provide empirical evidence and robustness tests showing that optimized strategy outperforms a baseline strategy in risk-adjusted performance.

Keywords: High-Frequency; Hamilton–Jacobi–Bellman (HJB) equation; Cryptocurrency; Triangular Arbitrage; Discrete dynamic programming (DP); Modigliani–Miller Theorem

JEL Classification: G12; G14; C61

1 Introduction

The Modigliani–Miller (M&M) theorem, first proposed by Modigliani and Miller (1958), established that under perfect market conditions, a firm’s value is independent of its capital structure. The theorem’s proof relies on arbitrage arguments: if two firms with identical cash flows but different capital structures have different market values, investors can create risk-free arbitrage profits by replicating leverage at the personal level. This paper shows that the same arbitrage mechanism can be applied to cryptocurrency contract markets, where alternative trading routes to acquire the same asset represent analogous “capital structures.”

Cryptocurrency contract markets provide an ideal setting for M&M-style multi-path arbitrage. Multiple trading pairs create both direct and indirect routes to the same asset; for example, XRP can be acquired directly via XRP/USDT or indirectly via XRP/ETH and ETH/USDT. In high-frequency trading, realized profitability is therefore highly sensitive to execution quality and path selection, while transaction costs, slippage, and latency remain binding constraints that can quickly eliminate apparent arbitrage profits (Almgren and Chriss, 2000; Hasbrouck, 2009; Hasbrouck and Saar, 2013; Makarov and Schoar, 2020). We address this by separating execution design from path selection: we trade through single-leverage contracts to improve executability under conservative all-taker fee assumptions, and we optimize trading paths using a network-based framework under transaction costs and slippage constraints without relying on price prediction. This framing complements the Efficient Market Hypothesis benchmark of Fama (1970) and related evidence that market frictions and inefficiencies can sustain exploitable deviations (Lo and MacKinlay, 1988; De Long *et al.*, 1990; Jegadeesh and Titman, 1993).

Our research question is: *Can an M&M-style multi-path arbitrage framework, combined with single-leverage execution and network-based path optimization, deliver stable within-exchange arbitrage performance under conservative transaction-cost assumptions?*

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This paper makes three contributions. First, drawing on the arbitrage logic of Modigliani and Miller (1958), we formalize trading routes as alternative capital structures, prove the equilibrium condition $V_1 = V_2$, and generalize it to $k + 1$ trading pairs. Second, we develop a network-based path-optimization approach for multi-path arbitrage under execution frictions without requiring price prediction, combining an HJB formulation of execution costs with discrete DP on the trading network, implemented as a Bellman recursion. Third, using high-frequency Binance data from January to December 2025 across 64 trading-pair groups, we present empirical results and robustness tests showing that the optimized strategy outperforms a baseline strategy in risk-adjusted performance.

The remainder of this paper is organized as follows. Section 2 reviews relevant literature and formulates empirical predictions. Section 3 presents the theoretical framework. Section 4 describes the empirical methodology, including data description and strategy implementation. Section 5 presents empirical results organized by empirical predictions. Section 6 presents robustness tests. Section 7 concludes by summarizing the main findings and discussing implications and future research directions.

2 Relevant Literature and Theoretical Implications

2.1 Related Literature

Prior research on arbitrage spans theoretical foundations, statistical methods, market microstructure, optimal control, and extreme market conditions. We synthesize these strands to motivate the empirical predictions in Subsection 2.2.

Theoretical Foundations: The no-arbitrage principle, which states that identical assets should have identical prices in efficient markets, was first formalized by Modigliani and Miller (1958) through the Modigliani-Miller theorem. This seminal work established that a firm’s value is independent of its capital structure under perfect market conditions, with the proof relying on arbitrage arguments: if two firms with identical cash flows but different leverage have different values, investors can create risk-free profits until prices converge. This arbitrage mechanism, where investors exploit temporary price deviations by going long undervalued and short overvalued positions, has been empirically validated in high-frequency markets (Stübinger, 2019). This established the fundamental no-arbitrage principle that underpins modern finance theory. Building on this foundation, Ross (1976) developed the Arbitrage Pricing Theory (APT) in 1976, modeling asset returns as linear functions of systematic risk factors, and Delbaen and Schachermayer (1994) provided a general version of the fundamental theorem of asset pricing, establishing rigorous mathematical foundations for no-arbitrage conditions. Further refinements include the second fundamental theorem in complete markets (Jarrow *et al.*, 1999), weak convergence links from discrete to continuous-time gain processes (Duffie and Protter, 1992), and extensions under proportional transaction costs in finite discrete time (Schachermayer, 2004). In our setting, no-arbitrage implies a path-based restriction: a direct route and an indirect route to the same asset should coincide in a frictionless benchmark. This motivates applying an M&M-style arbitrage logic to multi-path trading routes under transaction costs and execution constraints.

Statistical Arbitrage Methods: The application of statistical methods to arbitrage strategies was advanced by Gatev *et al.* (2006), who provided a rigorous framework for pairs trading based on mean reversion, showing that cointegrated assets exhibit predictable convergence behavior. This work established the theoretical basis for using statistical measures such as Z-scores to identify arbitrage opportunities, which directly informs our empirical strategy. Avellaneda and Lee (2010) further developed statistical arbitrage strategies using mean-reverting processes, modeling price

spreads as Ornstein-Uhlenbeck processes and establishing how to optimally time entry and exit points based on statistical thresholds. Recent work on statistical arbitrage with optimal causal paths on high-frequency data (Stübinger, 2019) extends these methods to path optimization. Research on robust statistical arbitrage strategies (Lütkebohmert and Sester, 2021) addresses performance stability concerns. Motivated by this literature, we adapt standard Z-score-based statistical-arbitrage tools to within-exchange cryptocurrency triangular arbitrage in contract markets, incorporating transaction costs and execution constraints.

Market Microstructure and Execution Costs: The role of market microstructure in arbitrage execution has been extensively examined. Roll (1984) developed a simple implicit measure of the effective bid-ask spread, providing a foundation for estimating transaction costs. Kyle (1985) examined continuous auctions and insider trading, modeling price formation mechanisms. Amihud (2002) analyzed illiquidity and stock returns, establishing cross-section and time-series effects that inform our slippage modeling approach. Almgren and Chriss (2000) developed optimal execution models for portfolio transactions, while Bouchaud *et al.* (2004) examined price impact and market microstructure effects. Hasbrouck (2009) showed that effective execution cost estimation requires modeling both permanent and temporary price impact, providing empirical evidence on trading costs and returns for U.S. equities. Hasbrouck and Saar (2013) examined low-latency trading, revealing that transaction costs and execution frictions create persistent arbitrage opportunities in fragmented markets. Makarov and Schoar (2020) documented significant price deviations across cryptocurrency exchanges, finding that arbitrage opportunities persist due to market fragmentation and trading frictions, with transaction costs being significant barriers to execution. Recent research by Koutmos *et al.* (2024) examined triangular arbitrage exploitability in cryptocurrency spot markets, finding that most opportunities are too small to overcome transaction costs, with only a tiny fraction being profitable even for VIP-level traders. Angstmann *et al.* (2021) analyzed slippage in automated market maker networks. Taken together, this literature indicates that the realizability of apparent arbitrage profits is highly sensitive to transaction costs and execution frictions; in our setting, we therefore incorporate explicit transaction-cost and slippage constraints into path optimization.

Optimal Control and Path Optimization: The application of optimal control theory to financial strategies was pioneered by Merton (1971), who proposed the optimal investment portfolio problem solved through the HJB equation. This framework provides a principled approach to optimizing trading strategies under uncertainty. Building on this core, network structure becomes central in our setting: Bonanno *et al.* (2001) introduced network analysis methods for understanding market structure and centrality, and Amini *et al.* (2016) provides evidence on how network structure governs resilience and shock propagation; utility maximization in large markets further clarifies how optimal decisions depend on market structure and admissible strategies (Mostovyi, 2018). Extensions under frictions and model uncertainty highlight how transaction costs alter dynamic control policies (Chau *et al.*, 2022). Research on relative arbitrage with sharp time horizons (Larsson and Ruf, 2021) and hedging under arbitrage constraints (Ruf, 2013) complements this control-theoretic view. These studies collectively provide the theoretical foundation for our network-based path optimization approach, which combines HJB theory with discrete DP.

Extreme Market Conditions and Robustness: The robustness of trading strategies under extreme market conditions has been examined by Taleb (2007), who emphasized risk management under unpredictable black swan events. Ang *et al.* (2006) examined downside risk and extreme volatility, while Christoffersen *et al.* (2011) explored portfolio optimization under model uncertainty. Research on robust statistical arbitrage strategies (Lütkebohmert and Sester, 2021) addresses performance stability under adverse conditions. Bailey *et al.* (2014) examined the probability of backtest overfitting, while White (2000) provided a reality check for data snooping,

both highlighting the importance of robust empirical validation. Collectively, these studies provide frameworks for designing robust strategies that can respond to extreme market volatility; we therefore evaluate strategy robustness under extreme market conditions and through comprehensive robustness tests.

2.2 Empirical Predictions

Based on the literature review, we formulate the following empirical predictions to be evaluated in Sections 5 and 6:

P1: Arbitrage opportunities can arise in cryptocurrency contract markets, and some triangular-arbitrage opportunities may remain economically exploitable after transaction costs; however, net profitability is expected to vary across asset groups due to differences in liquidity, execution frictions, and market structure.

P2: Market network structure significantly affects arbitrage profitability: strategies involving hub currencies are more likely to generate losses due to efficient price discovery, while those involving peripheral currencies show greater profitability potential.

P3: Network-based path optimization effectively improves arbitrage performance when applied to non-hub currencies with multiple viable paths.

3 Theoretical Framework

3.1 Base Case

$$V_1 = V_2$$

$$\text{or equivalently } P_{\text{direct}} = P_{\text{indirect}}, \quad \text{where } P_{\text{indirect}} := P_{\text{XRP/ETH}} \times P_{\text{ETH/USDT}}$$

Remark. V_1 and V_2 are the target asset’s numeraire prices implied by the direct route and by the indirect route, respectively, each constructed from quoted market prices (e.g., mid or close). They are route-specific *conversion rates*, not measures of fundamental or intrinsic value. Under frictionless no-arbitrage, the accounting identity $V_1 = V_2$ holds as a benchmark; *nominal* deviations $V_1 \neq V_2$ are the apparent mispricings that motivate spread-based trading signals. Transaction costs, slippage, and sequential execution risk are not part of this frictionless equality.

We consider two cases:

Case I: When $V_1 < V_2$

Suppose $V_1 < V_2$ (i.e., $P_{\text{direct}} < P_{\text{XRP/ETH}} \times P_{\text{ETH/USDT}}$). An investor holding α units of XRP via the indirect path (XRP/ETH) can execute the following arbitrage:

1. Sell α units of XRP/ETH, receiving $\alpha \cdot P_{\text{XRP/ETH}}$ units of ETH
2. Sell ETH for USDT, receiving $\alpha \cdot P_{\text{XRP/ETH}} \cdot P_{\text{ETH/USDT}}$ USDT
3. Use all funds ($\alpha \cdot V_2 = \alpha \cdot P_{\text{XRP/ETH}} \cdot P_{\text{ETH/USDT}}$) to buy XRP/USDT
4. Obtain $\alpha \cdot V_2/V_1$ units of XRP

The return from the new position is:

$$Y_1 = \left(\alpha \cdot \frac{V_2}{V_1} \right) \cdot \Delta_{\text{XRP}} = \alpha \cdot \Delta_{\text{XRP}} \cdot \frac{V_2}{V_1}$$

where α is the position size (in XRP units) and Δ_{XRP} is the XRP price change in USDT terms.

Since $V_2 > V_1$, we have $V_2/V_1 > 1$, so:

$$Y_1 = \alpha \cdot \Delta_{XRP} \cdot \frac{V_2}{V_1} > \alpha \cdot \Delta_{XRP}.$$

The right-hand side $\alpha \cdot \Delta_{XRP}$ is the return from simply holding the original α units of XRP. Therefore $Y_1 > \alpha \cdot \Delta_{XRP}$, and the arbitrage profit is

$$\text{Profit} = Y_1 - \alpha \cdot \Delta_{XRP} = \alpha \cdot \Delta_{XRP} \cdot \frac{V_2 - V_1}{V_1}.$$

This profit formula adapts Modigliani and Miller (1958): direct and indirect paths are analogous to capital structures, and profit arises from price deviations between them. We treat the intermediary's volatility σ_{ETH} as the "debt cost": as leverage has cost r in M&M, exposure to the intermediary over the execution window induces a risk cost in σ_{ETH} . Section 3.3 formalizes this.

Case II: When $V_1 > V_2$

The argument is symmetric: interchange the roles of the direct and indirect paths and reverse the trades in Case I. One obtains the analogous arbitrage profit

$$\text{Profit} = \alpha \cdot \Delta_{XRP} \cdot \frac{V_1 - V_2}{V_2}.$$

Since both cases yield strictly positive profit whenever $V_1 \neq V_2$, the no-arbitrage condition requires $V_1 = V_2$.

From benchmark to implementation.

Empirically, deviations from the benchmark $V_1 = V_2$ are monitored through the spread. We do *not* trade whenever $S_t \neq 0$; instead, we use rolling statistics over a lookback window W (minutes) and Z-score thresholds, as in Algorithm 1. Define

$$S_t = V_{2,t} - V_{1,t}. \tag{1}$$

Let μ_t and σ_t denote the rolling mean and standard deviation of S_t over $[t - W, t]$. The Z-score is

$$Z_t = \frac{S_t - \mu_t}{\sigma_t}. \tag{2}$$

Entry and exit are governed by

$$\begin{aligned} \text{entry: } & |Z_t| > k, \\ \text{exit: } & |Z_t| < \epsilon. \end{aligned} \tag{3}$$

Per-leg taker fees and dynamic slippage are applied at each entry and exit (Section 5.1).

3.2 Generalization to $k+1$ Trading Pairs

The above theorem establishes the equilibrium condition for the case of two trading paths, direct and indirect with one intermediary asset. We now generalize this result to $k+1$ trading pairs using mathematical induction.

Theorem (Generalized): Consider $k+1$ trading paths to acquire asset A in terms of base currency B :

- Direct path: $P_{A/B}$ (price of A in terms of B)
- k indirect paths: $P_{A/I_1} \times P_{I_1/B}, P_{A/I_2} \times P_{I_2/B}, \dots, P_{A/I_k} \times P_{I_k/B}$, where $I_1, I_2, \dots, I_k$ are intermediary assets

In the absence of arbitrage opportunities, all paths must have equal value:

$$\begin{aligned} P_{A/B} &= P_{A/I_1} \times P_{I_1/B} = P_{A/I_2} \times P_{I_2/B} \\ &= \dots = P_{A/I_k} \times P_{I_k/B} \end{aligned}$$

Proof by Mathematical Induction:

Base Case ($k = 1$): This is the case proven above, where we have:

$$P_{A/B} = P_{A/I_1} \times P_{I_1/B}$$

This establishes the base case.

Induction Hypothesis: Assume the theorem holds for k trading pairs. That is, for k intermediary assets $I_1, I_2, \dots, I_k$, we have:

$$\begin{aligned} P_{A/B} &= P_{A/I_1} \times P_{I_1/B} = P_{A/I_2} \times P_{I_2/B} \\ &= \dots = P_{A/I_k} \times P_{I_k/B} \end{aligned}$$

Induction Step: We now prove that if the theorem holds for k pairs, it also holds for $k+1$ pairs. Consider adding a $(k+1)$ -th intermediary asset I_{k+1} , creating a new indirect path $P_{A/I_{k+1}} \times P_{I_{k+1}/B}$.

By the induction hypothesis, we know that $P_{A/B} = P_{A/I_j} \times P_{I_j/B}$ for all $j \in \{1, 2, \dots, k\}$.

Now, consider the relationship between the direct path and the new $(k+1)$ -th indirect path. If $P_{A/B} \neq P_{A/I_{k+1}} \times P_{I_{k+1}/B}$, then without loss of generality, suppose $P_{A/B} < P_{A/I_{k+1}} \times P_{I_{k+1}/B}$.

An arbitrageur holding α units of A via the indirect path through I_{k+1} can execute the following strategy:

1. Sell α units of A for I_{k+1} , receiving $\alpha \cdot P_{A/I_{k+1}}$ units of I_{k+1}
2. Sell I_{k+1} for B , receiving $\alpha \cdot P_{A/I_{k+1}} \times P_{I_{k+1}/B}$ units of B
3. Use all funds, $\alpha \cdot P_{A/I_{k+1}} \times P_{I_{k+1}/B}$ units of B , to buy A via the direct path
4. Obtain $\alpha \cdot \frac{P_{A/I_{k+1}} \times P_{I_{k+1}/B}}{P_{A/B}}$ units of A

Since $P_{A/I_{k+1}} \times P_{I_{k+1}/B} > P_{A/B}$, the arbitrageur obtains more than α units of A , creating a risk-free profit. This contradicts the no-arbitrage assumption.

Similarly, if $P_{A/B} > P_{A/I_{k+1}} \times P_{I_{k+1}/B}$, the arbitrageur can reverse the strategy by selling via the direct path and buying via the indirect path to create a profit.

Therefore, the no-arbitrage condition requires:

$$P_{A/B} = P_{A/I_{k+1}} \times P_{I_{k+1}/B}$$

Combining this with the induction hypothesis, we have:

$$\begin{aligned} P_{A/B} &= P_{A/I_1} \times P_{I_1/B} = P_{A/I_2} \times P_{I_2/B} \\ &= \dots = P_{A/I_k} \times P_{I_k/B} = P_{A/I_{k+1}} \times P_{I_{k+1}/B} \end{aligned}$$

This completes the induction step. By mathematical induction, the theorem holds for any number of trading pairs $k + 1 \geq 2$.

This generalization shows that the M&M equilibrium condition extends beyond the specific case of two trading paths, one direct and one indirect, to any number of intermediary assets. It provides a robust theoretical foundation for multi-asset multi-path arbitrage strategies and for the path-optimization problem addressed by the network-based framework in Section 4.3.

3.3 Modeling Arbitrage Risk and Costs

3.3.1 Price-based Perspective

This subsection develops a price-based characterization of spread risk and execution cost, which serves as a theoretical reference and motivates the network-based approach in Section 3.3.2. We do not estimate or use κ_S , θ , or γ in our implementation; Section 3.3.2 then shows how network structure can substitute for the price-based measures above. Readers primarily interested in the practical implementation may proceed directly to Section 3.3.2.

Spread Process Modeling: The spread $S_t = V_{2,t} - V_{1,t}$ represents the deviation from the no-arbitrage equilibrium condition. Under the M&M framework, in equilibrium we have $V_1 = V_2$, so $S_t = 0$. When markets are inefficient, the spread follows a mean-reverting process with stochastic volatility:

$$dS_t = \kappa_S(\theta - S_t)dt + \sigma_S(S_t, t)dW_t \quad (4)$$

where $S_t = V_{2,t} - V_{1,t}$ is the spread, $\kappa_S > 0$ is the mean-reversion speed, θ is the long-term mean, typically zero under no-arbitrage, $\sigma_S(S_t, t)$ is the state-dependent volatility of the spread, and W_t is a standard Brownian motion. The mean-reversion term $\kappa_S(\theta - S_t)$ captures the tendency of the spread to return to equilibrium, while the diffusion term captures the uncertainty in convergence timing.

Spread Divergence Risk: The risk in spread trading arises from the possibility that the spread may diverge further before converging, or may not converge at all. This risk is quantified by the variance of the spread process. Using Itô's lemma, the instantaneous variance of the spread is:

$$\sigma_S^2 = \text{Var}(dS_t) = \sigma_S^2(S_t, t)dt \quad (5)$$

For a spread trading position with position size α , the risk exposure depends on the holding period t and the spread volatility. The expected risk cost over the holding period is:

$$\text{Risk Cost} = \frac{\gamma}{2} \cdot \alpha^2 \cdot \sigma_S^2 \cdot t \quad (6)$$

where γ is the risk aversion coefficient, α is the position size, and t is the expected holding period until convergence.

Intermediary Asset Volatility Impact: The spread volatility σ_S depends on the volatilities of V_1 and V_2 . Since $V_2 = P_{\text{XRP/ETH}} \times P_{\text{ETH/USDT}}$ involves the intermediary asset ETH, the spread volatility incorporates the volatility of the intermediary asset. Using the variance decomposition:

$$\sigma_S^2 = \sigma_{V_1}^2 + \sigma_{V_2}^2 - 2\rho_{V_1, V_2}\sigma_{V_1}\sigma_{V_2} \quad (7)$$

where σ_{V_1} and σ_{V_2} are the volatilities of V_1 and V_2 , and ρ_{V_1, V_2} is their correlation. The indirect path V_2 introduces additional volatility from the intermediary asset, which increases the spread volatility and thus the risk of divergence.

Execution timing and intermediary risk: The foregoing assumes that spread positions can be opened and closed at a single point in time. When orders cannot be executed instantaneously, for example when the legs of a triangular trade are filled sequentially, the arbitrageur is exposed to the intermediary asset ETH over the execution window. This exposure induces an additional risk cost: the variance of the intermediary's return over that window, which is driven by σ_{ETH} and the length of the lag. In our notation, this corresponds to an "ETH risk cost" or, more generally, an intermediary execution risk. It reduces the realized profit from the frictionless benchmark $\alpha \cdot \Delta_{XRP} \cdot (V_2 - V_1)/V_1$ derived in Section 3 and is captured in the spread variance (5) and its decomposition (7) through σ_{V_2} , which incorporates the intermediary's volatility.

The framework integrates two key components: spread divergence risk modeling, which quantifies the risk that the spread may not converge or may diverge further, and transaction cost estimation, including both fixed trading fees and dynamic slippage. The foregoing risk measures are expressed in terms of prices and volatilities. In Section 3.3.3 we explain why certain properties of the network structure can substitute for these price-based indicators.

3.3.2 Network-based Perspective

We now extend the risk and cost framework to incorporate stochastic volatility and optimal execution control using the HJB equation; in the discrete path-optimization below we use only network structure, and Section 3.3.3 explains why it can substitute for the price-based risk indicators of Section 3.3.

Problem Setup. Consider the problem of selling Q units of a digital asset within the time interval $[0, T]$. The asset price P follows a jump-diffusion process:

$$dP_t = \mu P_t dt + \sigma_P P_t dW_t + J P_t dN_t$$

where μ is the drift, σ_P is the price volatility, W_t is a standard Brownian motion (note: distinct from lookback window W), N_t is a Poisson process with intensity λ , and the jump amplitude $J \sim \text{Uniform}(-0.5\%, 0.5\%)$. Instantaneous selling quantity q_t linearly impacts the price:

$$P_t \rightarrow P_t(1 - \kappa q_t)$$

where q_t is the instantaneous trade rate and κ is the market impact coefficient. Each transaction incurs a fixed cost c and a proportional cost $\phi q_t P_t$. The optimization is subject to the constraints:

$$\int_0^T q_t dt = Q, \quad q_t \geq 0$$

where Q is the total quantity to execute and T is the time horizon.

Solution. Design a stochastic control model to solve for the optimal selling rate $\{q_t^*\}$ that minimizes the expected total cost:

$$\min_{\{q_t\}} E \left[\int_0^T (c \cdot \mathbf{1}_{\{q_t > 0\}} + \phi q_t P_t) dt + \kappa \int_0^T q_t P_t dt \right]$$

Define the value function as:

$$V(t, Q, P) = \min_{\{q_s\}_{s \geq t}} E_t \left[\int_t^T (c \cdot \mathbf{1}_{\{q_s > 0\}} + \phi q_s P_s) ds + \kappa \int_t^T q_s P_s ds \mid Q_t = Q, P_t = P \right]$$

The state dynamics are:

$$dQ_t = -q_t dt \tag{8}$$

$$dP_t = \mu P_t dt + \sigma P_t dW_t + J P_t dN_t \tag{9}$$

By the DP principle, the HJB equation is:

$$0 = \min_{q \geq 0} \left\{ c \cdot \mathbf{1}_{\{q > 0\}} + \phi q P + \kappa q P - q V_Q + V_t + \mu P V_P + \frac{1}{2} \sigma^2 P^2 V_{PP} + \lambda E_J [V(t, Q, P(1 + J)) - V(t, Q, P)] \right\}$$

Boundary conditions:

$$V(T, Q, P) = 0, \quad V(t, 0, P) = 0$$

When $\lambda = 0$, the HJB equation reduces to:

$$0 = \min_{q \geq 0} \left\{ c \cdot \mathbf{1}_{\{q > 0\}} + \phi q P + \kappa q P - q V_Q + V_t + \mu P V_P + \frac{1}{2} \sigma^2 P^2 V_{PP} \right\}$$

If we ignore the price sensitivity of the value function (V_P , V_{PP}) and approximate the price as constant, the optimal strategy is approximately uniform liquidation:

$$q_t^* = \frac{Q}{T}$$

When $\lambda > 0$, jump risk introduces jump integral terms in the value function and can lead to acceleration or batch liquidation strategies; for our short-horizon arbitrage path optimization we take $\lambda \approx 0$, as justified in the discrete DP approach below.

Why $\lambda \approx 0$: benchmark role of the jump-augmented diffusion.

We start from a jump-diffusion for P_t because it nests the standard continuous-time execution benchmark used in the optimal-liquidation literature: Brownian noise plus an optional Poisson jump component for rare discrete price moves. Our empirical implementation does *not* estimate a jump intensity from high-frequency returns, nor does it solve the HJB partial differential equation numerically. Instead, the discrete DP routine in Equation 10 and Algorithm 2 is motivated by the $\lambda \rightarrow 0$ *limit* of this benchmark: short-horizon, leg-by-leg cost aggregation does not require modeling jump-driven liquidation dynamics, while retaining additivity of per-leg fees, impact, and penalties as the economically relevant structure.

The uniform liquidation result suggests that, in this control-theoretic setup, the requirement for immediate execution is not particularly high; the discrete path optimization and empirical strategies below build on this intuition.

Discretization for Path Optimization. The continuous HJB equation framework provides theoretical motivation, but for practical implementation in cryptocurrency path optimization, we employ a discrete DP approach. This discrete implementation is consistent with weak convergence results linking discrete-time gain processes to continuous-time limits (Duffie and Protter, 1992). This simplification is justified because (i) price changes within an arbitrage path are small, allowing us to ignore price sensitivity terms (V_P , V_{PP}); (ii) jump risk is negligible for short-duration arbitrage cycles ($\lambda \approx 0$); and (iii) each trading leg represents a discrete time step, making discrete DP more appropriate than continuous HJB.

The discrete value function for path optimization is:

$$V(t, Q) = \min_{\text{leg}} \left[c \cdot \mathbf{1}_{\{\text{leg} > 0\}} + (\phi + \kappa(\text{leg})) \cdot Q \cdot P + V(t + 1, Q - \text{leg}) \right] \quad (10)$$

where c is the fixed transaction cost per leg, ϕ is the proportional transaction cost, $\kappa(\text{leg})$ is the market impact coefficient estimated from network-based analysis, Q is the remaining trade quantity, equal to 1 for arbitrage, and P is the price, assumed constant as we only care about relative costs.

We use breadth-first search combined with DP to find the path that minimizes total cost. The algorithm explores all possible paths from the base currency to the target currency, typically USDT, and selects the path with minimum penalty score:

$$\text{Penalty Score} = \sum_{e \in \text{path}} [\text{Transaction Cost} + \text{Slippage Estimate}(e)] + \text{Network Centrality Penalty}$$

The network centrality penalty accounts for market efficiency: currencies with higher centrality, i.e., more trading connections, typically have lower arbitrage opportunities due to market efficiency.

3.3.3 Why Network Structure Can Substitute for Price-Based Risk Measures

Section 3.3 quantified risk using price and volatility data and noted that we would explain why network properties can substitute for them. The path cost and penalty score above use only network structure: path length, centrality, and connectivity. We posit that these proxy for the three risk dimensions in Section 3.3.

(i) Spread divergence risk. Spread divergence is partly driven by path length and intermediary fragility; the path-length penalty and connectivity in the network capture this.

(ii) Intermediary volatility. Intermediary volatility (σ_{ETH} , σ_{V_2}) is correlated with centrality: hubs tend to be more liquid and stable, peripheries more volatile; the centrality-based terms in $\kappa(\text{leg})$ and the network centrality penalty proxy for this.

(iii) Execution-timing risk. Execution-timing risk rises with the number of legs and trading difficulty; path length and $\kappa(\text{leg})$ as a network-based slippage proxy reflect it.

Thus, the penalty score and $\kappa(\text{leg})$ substitute for Section 3.3’s price-based measures when such data are unavailable. The specific functional forms and calibrated parameters for the slippage estimation and centrality penalty are detailed in Section 4.3.

4 Empirical Methodology

4.1 Data Description

We use minute-level OHLCV (Open, High, Low, Close, Volume) data from Binance exchange for the period January 1 to December 31, 2025. Our analysis covers 64 trading pair groups for strategy S_1 and 64 trading pair groups for strategy S_2 .

Each group forms a triangular arbitrage structure. For example, Group 1 (XRP) consists of:

- Direct path: XRPUSDT
- Indirect path: XRPETH and ETHUSDT

4.2 Baseline Strategy (S_1)

We implement the baseline triangular arbitrage strategy S_1 across 64 cryptocurrency groups to examine exploitable arbitrage opportunities and performance stability. Both S_1 and S_2 are implemented as *spread trading* strategies: rather than executing three sequential cryptocurrency trades in a round-trip, a position is opened simultaneously on both path values V_1 and V_2 —long one and short the other—to profit from spread convergence. Specifically, going long V_1 means buying the direct pair A/B , while going short V_2 means selling the indirect pair legs A/I and I/B ; the reverse applies for the other direction. Transaction costs of 0.05% per leg (three legs total, all at taker rates) and dynamic slippage are applied at each entry and exit, with adverse slippage applied separately to each path value V_1 and V_2 using Equation 14.

The Z-score signal governs entry and exit: a position is opened when the Z-score of the spread exceeds the entry threshold k , and closed when it falls below the exit threshold ϵ .

Algorithm 1 presents the complete backtesting procedure.

Algorithm 1 Baseline Triangular Arbitrage Strategy (S_1)

Require:

- 1: OHLCV data for trading pairs: $P_{direct}, P_{leg1}, P_{leg2}$
- 2: Parameters: lookback window $W = 60$ min, entry threshold $k = 3.0$, exit threshold $\epsilon = 0.5$
- 3: Transaction cost $f = 0.05\%$ per leg, initial capital C_0

Ensure:

- 4: Portfolio value series $\{C_t\}$ for performance evaluation
 - 5:
 - 6: Initialize: $C_t \leftarrow C_0, position \leftarrow 0$
 - 7: **for** each timestamp t **do**
 - 8: Compute spread: $S_t \leftarrow P_{leg1,t} \times P_{leg2,t} - P_{direct,t}$
 - 9: Compute rolling statistics: μ_t, σ_t over window $[t - W, t]$
 - 10: Compute Z-score: $Z_t \leftarrow (S_t - \mu_t) / \sigma_t$
 - 11: **if** $position = 0$ AND $|Z_t| > k$ **then**
 - 12: $direction \leftarrow \text{sign}(Z_t) \{+1: \text{long } V_1 \text{ short } V_2; -1: \text{reverse}\}$
 - 13: Estimate slippage: $\kappa_{V_1}, \kappa_{V_2}$ using Equation 14
 - 14: Execute entry with transaction cost f and slippage κ
 - 15: $position \leftarrow direction, entry_price \leftarrow P_{direct,t}$
 - 16: **else if** $position \neq 0$ AND $|Z_t| < \epsilon$ **then**
 - 17: Estimate exit slippage: $\kappa'_{V_1}, \kappa'_{V_2}$
 - 18: Execute exit with transaction cost f and slippage κ'
 - 19: Update portfolio: $C_t \leftarrow C_t + \text{realized_pnl} - \text{exit_cost}$
 - 20: $position \leftarrow 0$
 - 21: **end if**
 - 22: **end for**
 - 23: **return** $\{C_t\}$
-

4.3 Path Optimized: Enhanced Strategy (S_2)

Building on S_1 , we develop enhanced strategy S_2 that replaces predetermined routes with path selection based on the discrete Bellman recursion in Section 3.3.2 (Equation 10 and Algorithm 2), which implements the additive execution-cost structure motivated by the continuous HJB formulation without numerically solving the HJB PDE. S_1 uses fixed paths. In S_2 , prior to backtesting, we apply this DP-based path optimization to all 64 cryptocurrency groups, regardless of their performance under S_1 , using only the network cost model to identify optimal paths. We retain the starting and target currencies of each predetermined path and minimize total path cost between them, allowing a variable number of intermediate currencies. The selected paths are then used in the subsequent backtest. This design allows us to evaluate whether DP-based path optimization improves outcomes across diverse market conditions and asset combinations.

Trading Network Structure. We collect 1,315 tradable pairs from the full market and represent the trading network as a directed graph $G = (V, E)$, where V is the set of currencies and E is the set of tradable pairs; an edge between two currencies indicates that one can transfer between them via that pair (see Figure 1). The network exhibits a core-periphery structure. On closer inspection, currencies with many incident edges appear as densely connected nodes; these are hub currencies, through which a large share of exchange flows. Other currencies are more likely to be converted into hub currencies, while hub currencies are relatively more likely to be held, creating a supply-demand asymmetry intuition. This asymmetry is the economic rationale for our hub-periphery distinction. Peripheral currencies are those with fewer connections, occupying peripheral

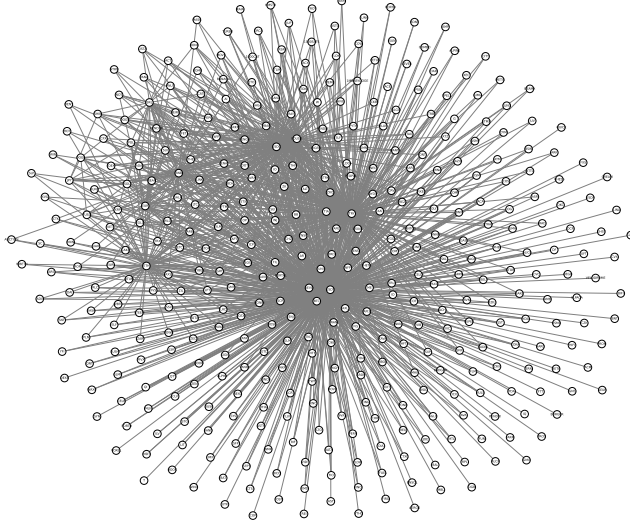


Figure 1: Cryptocurrency Trading Network. Hub currencies (more connections) are positioned centrally; peripheral currencies (fewer connections) are positioned peripherally. This structure determines the centrality scores used in the path optimization penalty function.

positions.

This structural distinction has important implications for arbitrage profitability. Hub currencies such as ETH, BTC, and BNB tend to exhibit smaller arbitrage opportunities due to efficient price discovery. Their high connectivity facilitates rapid information diffusion and price convergence. In contrast, peripheral currencies tend to maintain more stable profitability, as their lower connectivity and trading volume result in pricing inefficiencies that create exploitable arbitrage opportunities. Figure 1 illustrates this structure.

Centrality Calculation. To quantify the hub/peripheral distinction, we calculate degree centrality for each currency $v \in V$ based on network structure. Let $d(v)$ denote the degree of currency v , i.e. the number of connections, and $f(v)$ denote its frequency, i.e. the number of trading pairs involving v . We define a composite centrality score:

$$C(v) = \omega_f \cdot \frac{f(v)}{\max_{u \in V} f(u)} + \omega_d \cdot \frac{d(v)}{\max_{u \in V} d(u)} \quad (11)$$

where ω_f and ω_d are weights on frequency and degree ($\omega_f + \omega_d = 1$). We set $\omega_f = 0.7$ and $\omega_d = 0.3$ so that trading frequency has greater weight than connectivity. Currencies are classified as hub currencies if their centrality scores rank in the top 12. Parameter sensitivity analysis for centrality weights is reported in Table B10 in the Appendix.

Path Cost Model. We implement Equation 10—the discrete Bellman recursion motivated in Section 3.3.2—as a DP problem on the trading network $G = (V, E)$. Unlike S_1 ’s price-based slippage model (Equation 14), S_2 estimates market impact using only network structure.

Algorithm 2 Network Path Optimization via DP (S_2)

Require:

- 1: Trading network $G = (V, E)$ (1,315 Binance trading pairs)
- 2: Start currency s (e.g., XRP), target currency t (e.g., USDT)
- 3: Centrality scores C (from Equation 11)
- 4: Maximum path length $L_{\max} = k$
- 5: κ (Eq. 12): $\text{BASE} = 0.01\%$, $\alpha_{\text{edge}} = 0.10\%$, $\alpha_{\text{start}} = \alpha_{\text{end}} = 0.05\%$

Ensure:

- 6: Optimal path $P^* = [s, c_1, \dots, c_k, t]$ minimizing Equation 13
 - 7: Total path cost $\text{Cost}(P^*)$
 - 8:
 - 9: Initialize: $DP[(s, 0)] \leftarrow (0, [s])$
 - 10: $\text{best_cost} \leftarrow \infty$, $\text{best_path} \leftarrow \emptyset$
 - 11: **for** $\ell = 2$ **to** $L_{\max}$ **do**
 - 12: **for** each state $(v, k) \in DP$ where $k = \ell - 1$ **do**
 - 13: **for** each neighbor $u \in N(v)$ in graph G **do**
 - 14: **if** u in path **then**
 - 15: **continue** {Avoid cycles}
 - 16: **end if**
 - 17: Compute edge cost: $\text{cost}_{\text{edge}} = f + \phi + \kappa(v, u)$ {Eq. 12}
 - 18: $\text{cost}_{\text{total}} \leftarrow DP[(v, k)].\text{cost} + \text{cost}_{\text{edge}}$
 - 19: **if** $(u, k+1) \notin DP$ OR $\text{cost}_{\text{total}} < DP[(u, k+1)].\text{cost}$ **then**
 - 20: $DP[(u, k+1)] \leftarrow (\text{cost}_{\text{total}}, DP[(v, k)].\text{path} \cup \{u\})$
 - 21: **end if**
 - 22: **if** $u = t$ AND $k+1 = \ell$ **then**
 - 23: $\text{cost}_{\text{final}} \leftarrow \text{cost}_{\text{total}} + \lambda_{\text{len}} \times (\ell - 2)$ {Length penalty}
 - 24: **if** $\text{cost}_{\text{final}} < \text{best_cost}$ **then**
 - 25: $\text{best_cost} \leftarrow \text{cost}_{\text{final}}$
 - 26: $\text{best_path} \leftarrow DP[(v, k)].\text{path} \cup \{u\}$
 - 27: **end if**
 - 28: **end if**
 - 29: **end for**
 - 30: **end for**
 - 31: **if** $\ell \in \{2, 3\}$ AND $\text{best_cost} < 0.01$ **then**
 - 32: **break** {Early stop if short path is efficient}
 - 33: **end if**
 - 34: **end for**
 - 35: **return** $\text{best_path}, \text{best_cost}$
-

For each edge $(v, u) \in E$, the network-based slippage coefficient $\kappa(v, u)$ is defined as:

$$\kappa(v, u) = \text{BASE} + \alpha_{\text{edge}} \cdot C_{\text{edge}}(v, u) + \alpha_{\text{start}} \cdot C_{\text{start}} + \alpha_{\text{end}} \cdot C_{\text{end}} \quad (12)$$

where:

- $C_{\text{edge}}(v, u) = [C(v) + C(u)]/2$ is the edge centrality (average centrality of the two currencies);
- $C_{\text{start}} = C(s)$ if the path start currency s is a hub currency, 0 otherwise;
- $C_{\text{end}} = C(t)$ if the path end currency t is a hub currency, 0 otherwise.

This $\kappa(v, u)$ serves as the network-based substitute for the price-based risk and execution terms discussed in Section 3.3 and Section 3.3.2. Parameter sensitivity analysis for the κ parameters (BASE, α_{edge} , α_{start}) is reported in Tables B11, B12, and B13 in the Appendix.

For each path $P = [c_0, c_1, \dots, c_k]$, the total cost is:

$$\text{Cost}(P) = \sum_{i=0}^{k-1} [f + \phi + \kappa(c_i, c_{i+1})] + \sum_{j=1}^{k-1} \alpha_{\text{intermediate}} \cdot C(c_j) \cdot \mathbf{1}_{\{c_j \in \mathcal{H}\}} + \lambda_{\text{len}} \times (k-1) \cdot \mathbf{1}_{\{k > 2\}} \quad (13)$$

where $f = 0.05\%$ is the fixed transaction cost per leg, $\phi = 0.05\%$ is the proportional cost, i.e., the taker fee, $\kappa(c_i, c_{i+1})$ is the network-based market impact from Equation 12, $\alpha_{\text{intermediate}} = 0.08\%$ is the intermediate node penalty coefficient, $C(c_j)$ is the centrality score of currency c_j , $\mathcal{H}$ is the set of hub currencies, $\mathbf{1}_{\{\cdot\}}$ is the indicator function, and $\lambda_{\text{len}} = 0.0001$ is the path length penalty coefficient (distinct from the Poisson jump intensity λ in the HJB setup in Section 3.3.2). The intermediate node penalty term $\sum_{j=1}^{k-1} \alpha_{\text{intermediate}} \cdot C(c_j) \cdot \mathbf{1}_{\{c_j \in \mathcal{H}\}}$ penalizes paths that pass through hub currencies as intermediate nodes, reflecting the higher market impact and lower arbitrage opportunities associated with hub currencies. The term $\lambda_{\text{len}} \times (k-1) \cdot \mathbf{1}_{\{k > 2\}}$ penalizes paths with more than 2 trading pairs (i.e., $k > 2$), as longer paths accumulate transaction costs and slippage. Algorithm 2 searches over paths with 2 to k trading pairs ($L_{\text{max}} = k$, set in implementation) to minimize Equation 13.

5 Empirical Results

5.1 Dynamic Slippage Model

Within the backtest, we simulate execution by applying a dynamic slippage model at each entry and exit to approximate realistic fill prices for both S_1 and S_2 ; the resulting prices are used to compute P&L. Following Almgren and Chriss (2000), Hasbrouck (2009), and Roll (1984), the model uses observable OHLCV data and combines Almgren-Chriss market impact, an effective bid-ask spread proxy, and an illiquidity measure:

$$\begin{aligned} \text{Slippage}(t, Q, P, V) = & \text{BASE} + w_1 \times \text{Slippage}_{\text{AC}}(Q, V, \sigma) \\ & + w_2 \times \text{Effective Spread}(P) + w_3 \times \text{Illiquidity}(P, V, Q) \end{aligned} \quad (14)$$

where $\text{Slippage}_{\text{AC}} = \eta \times (Q/V)^\beta \times \sigma^\gamma$ follows the Almgren-Chriss framework with square root law (Bouchaud *et al.*, 2004), $\text{Effective Spread} = (P_{\text{max}} - P_{\text{min}})/\text{Mid}$ captures implicit bid-ask spread, and $\text{Illiquidity} = |\text{Return}|/\text{Volume}$ measures price impact per unit volume. Here Q is trade size, P is price, V is volume, σ is volatility, and w_1, w_2, w_3 are weights on the three slippage components. Parameters are calibrated for high-liquidity cryptocurrency markets: $\text{BASE} = 0.01\%$, $\eta = 0.05$, $\beta = 0.5$, $\gamma = 0.25$, with slippage clipped to $[0.01\%, 0.2\%]$. Parameter sensitivity analysis for w_1 is reported in Table B14 in the Appendix.

5.2 Strategy S_1 Results

The results show that S_1 consistently generates positive returns for selected asset combinations, confirming that arbitrage opportunities exist and can be systematically exploited despite transaction costs and execution constraints. However, S_1 's performance is not stable across all market conditions: while some cryptocurrency groups show stable positive returns, others show mixed results or consistent losses. This validates empirical prediction P1, which posits that arbitrage opportunities exist in cryptocurrency markets that can be systematically exploited, but the baseline strategy shows unstable performance across different asset combinations. Table B1 (see Appendix) presents the complete performance statistics for all 64 groups using the baseline triangular arbitrage strategy S_1 , and figure 2 presents the cumulative returns.

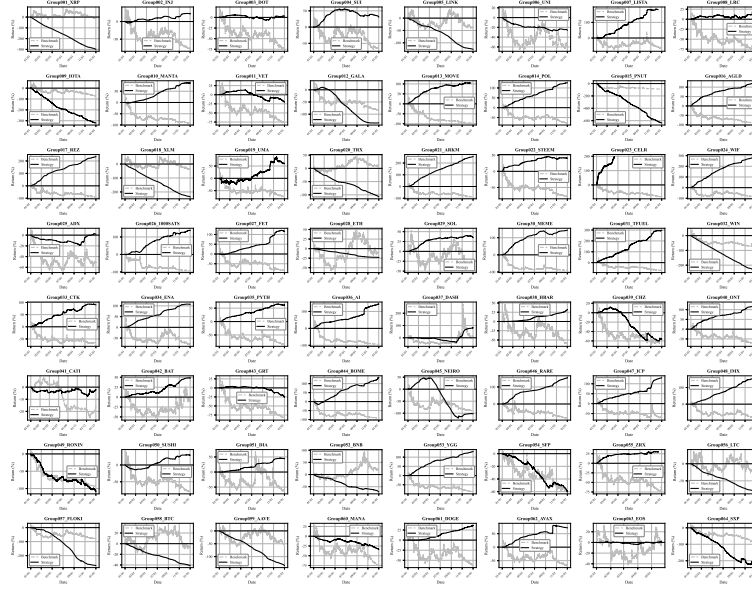


Figure 2: S_1 Backtest Returns. This figure presents the backtest returns for S_1 , with buy-and-hold strategy as the benchmark. The benchmark holds the starting currency of the triangular arbitrage. Complete performance statistics for all 64 groups are provided in Table B1 in the Appendix.

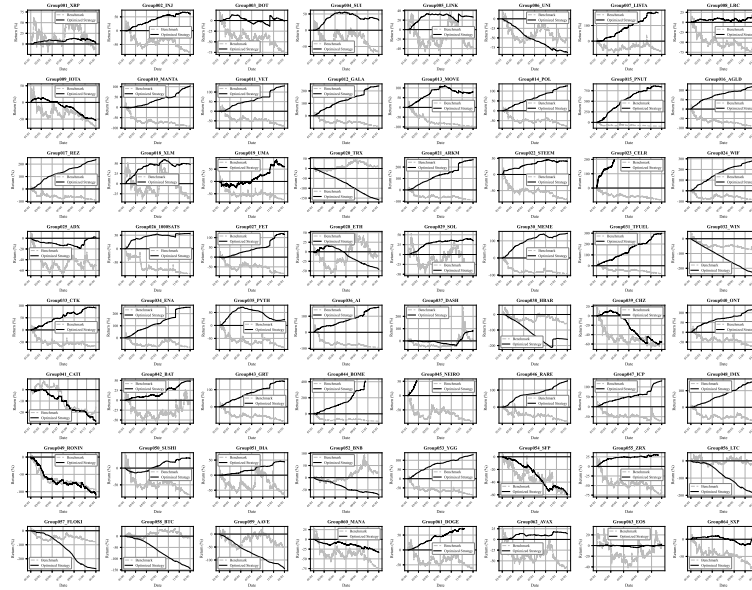


Figure 3: S_2 Backtest Returns. This figure presents the backtest returns for optimized strategy S_2 (network-based path selection as in Subsection 4.3), with all other parameters identical to S_1 . Complete performance statistics for all 64 groups are provided in Table A2 in the Appendix.

5.3 Strategy S_2 Results

Table B2 (see Appendix) presents complete performance statistics for all 64 groups under optimized strategy S_2 , and figure 3 presents the cumulative returns.

The path optimization algorithm systematically explored paths with 2 to k trading pairs for each of the 64 cryptocurrency groups. Across all 64 groups, 2-pair triangular arbitrage paths were consistently selected as optimal, as longer paths' accumulated transaction fees of 0.05% per leg and slippage outweighed their route diversity benefits. This empirical finding validates the practical efficiency of triangular arbitrage while confirming the applicability of our theoretical framework to path optimization.

A notable finding is that a subset of strategies continues to exhibit losses; these are precisely those whose *start* currency is a hub, such as BTC, ETH, and LTC. This aligns with the economic intuition underlying our network design: hub currencies are more likely to generate losses due to efficient price discovery, while peripheral currencies offer greater profitability. Under S_2 , even when selecting the optimal path over all candidates, we observe that paths starting with hub currencies generally fail to yield positive returns in our sample. These findings suggest that implementations of S_2 -style path optimization should prioritize peripheral currencies as the start currency and stablecoins such as USDT or USDC as the terminal anchor currency. The anchor currency is the asset in which the terminal position is held and, under our buy-and-hold benchmark, is not rebalanced; under other benchmarks, a different anchor could be used. Collectively, the S_2 backtest results support empirical prediction P2 on hub versus peripheral effects on profitability and empirical prediction P3 on the effectiveness of network-based path optimization for non-hub currencies with multiple viable paths.

5.4 Comparison of S_1 and S_2

The comparison between S_1 and S_2 , as shown in Table B1 and Table B2 in the Appendix, shows that network-based path optimization improves risk-adjusted returns. For groups that already showed positive returns under S_1 , S_2 either maintained or further improved performance, demonstrating the broad applicability of the path optimization framework.

The empirical results reveal that strategy S_1 , despite employing random path selection, generated positive returns for 64.0625% of all groups, while optimized strategy S_2 elevated this proportion to 75.0000%. Nevertheless, the principal contribution of S_2 is not simply to boost the aggregate positive return rate, but to systematically evaluate whether path optimization can forestall negative returns, preserve positive returns, or augment returns. Following this evaluation framework, we categorize results as *optimized* if S_2 returns are greater than or equal to S_1 returns, and as *failed* if S_2 returns fall below S_1 returns.

Our analysis identifies two distinct categories that necessitate specialized treatment. Currencies with connection points ≤ 2 have a unique path configuration, thereby excluding the possibility of DP-based path pre-screening. These instances, classified as *No Additional Path*, are assessed solely via extensive historical backtesting. Furthermore, we observe a subset of currencies characterized by high global centrality metrics yet demonstrating persistent losses across both strategies; these are categorized as *Hub* currencies.

Figure 4 shows hierarchical clustering of the 64 strategies by similarity of their cumulative return time series. Note that this figure refers to *cross-sectional* clustering of strategies by similarity of cumulative return paths (dendrogram based on pairwise correlations); it is not used to identify a self-exciting jump arrival process for the midprice P_t , nor does it speak directly to Poisson versus Hawkes modeling of jumps. For S_1 , many strategies exhibit similar return dynamics, with a large number converging to -100% returns, yielding a relatively homogeneous clustering pattern. For S_2 ,

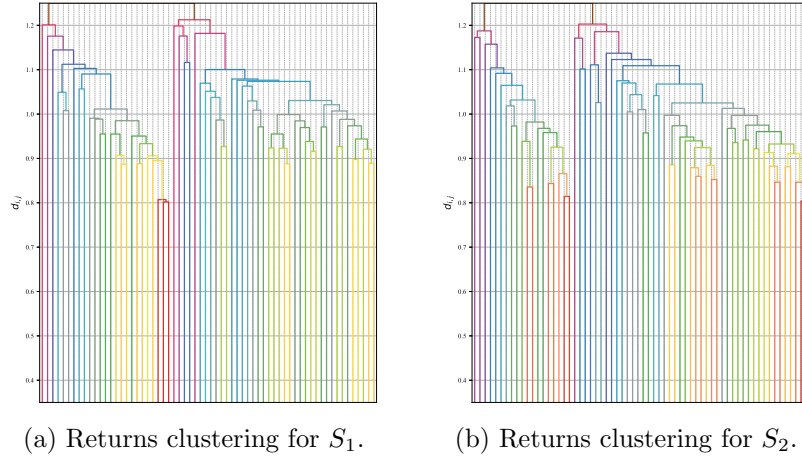


Figure 4: Returns clustering for S_1 and S_2 . Hierarchical clustering of the 64 strategies by similarity of cumulative return time series: pairwise distance $(1 - \rho)/2$ with ρ the Pearson correlation between return paths; average linkage. Vertical axis: merge dissimilarity $d_{i,j}$. Link color: merge-level dissimilarity (diverging scale). Horizontal axis: strategies in dendrogram leaf order. Panel (a): S_1 ; (b): S_2 . The greater spread in merge heights and link colors in (b) reflects more heterogeneous return dynamics under S_2 , consistent with path optimization affecting asset combinations differently.

the dendrogram is more differentiated: merges occur over a wider range of heights and link colors show greater spread, reflecting more varied return distributions across strategies. This increased heterogeneity is consistent with P3: network-based path optimization improves performance differentially across asset combinations, with greater improvements for non-hub currencies with multiple paths and smaller improvements for hub or single-path cases, resulting in more diverse return dynamics under S_2 compared to S_1 , as shown in Table B1 and Table B2. This validates empirical prediction P3, demonstrating that optimized strategy S_2 can substantially improve performance, particularly for non-hub currencies with multiple trading paths.

6 Robustness Tests

The robustness of trading strategies under various market conditions and methodological assumptions is crucial for practical implementation. Following the literature on extreme events, overfitting, and data snooping (Taleb, 2007; Ang *et al.*, 2006; Christoffersen *et al.*, 2011; Lütkebohmert and Sester, 2021; Bailey *et al.*, 2014; White, 2000), we conduct five comprehensive robustness tests to validate the reliability and practical applicability of our findings, addressing key concerns in empirical asset pricing: statistical significance, economic viability, parameter sensitivity, temporal stability, and sample-specific bias. These tests are designed to ensure our results represent genuine arbitrage opportunities rather than artifacts of random variation, methodological choices, or data characteristics.



Figure 5: Bootstrap Statistical Validation. All 64 strategies: y -axis = strategy, sorted by bootstrap mean Sharpe; x -axis = bootstrap mean Sharpe with 95% confidence intervals shown as horizontal error bars. Filled circles = statistically significant at the 5% level; open circles = non-significant at the 5% level.

6.1 Bootstrap Resampling

We conduct bootstrap resampling with 1000 iterations for all 64 strategies to assess the statistical significance of observed Sharpe ratios. For each strategy, we resample daily returns with replacement, compute Sharpe ratios for each bootstrap sample, and construct 95% confidence intervals. Strategies with confidence intervals excluding zero are considered statistically significant at the 5% level. Figure 5 presents the results. High-performing strategies such as ENA, POL, REZ, PNUT, IMX show strong significance at $p < 0.001$; hub-currency strategies such as ETH, BTC, and LTC show significantly negative returns. The bootstrap analysis confirms that successful arbitrage strategies generate statistically significant excess returns rather than measurement noise. Complete results are in Table B3 in the Appendix.

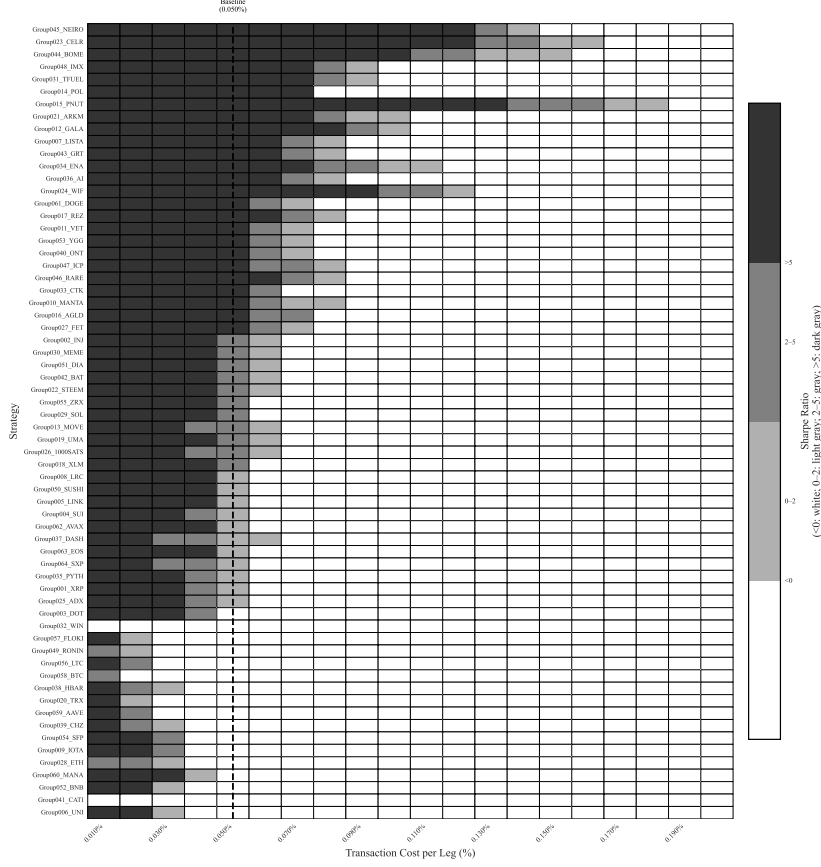


Figure 6: Transaction Cost Sensitivity. Heatmap of all 64 strategies by 20 transaction cost levels per leg, 0.01%–0.20%. Rows: strategies; columns: cost levels. Shading = Sharpe ratio; vertical dashed line = baseline 0.05%.

6.2 Transaction Cost Sensitivity

Transaction costs critically impact arbitrage profitability. We test strategy performance across 20 cost levels from 0.01% to 0.20% per leg, with baseline 0.05%, a conservative full taker fee in contract markets. Figure 6 is a heatmap: the y -axis shows all 64 strategies, sorted by Sharpe at a cost nearest to 0.05%; the x -axis shows the 20 cost levels; cell shading indicates Sharpe ratio on a diverging scale centered at zero. A vertical dashed line marks the baseline 0.05%. High-resilience strategies remain in positive-Sharpe regions to higher cost levels; low-resilience ones turn negative near or below baseline. The analysis shows that 47 of 64 strategies, or 73%, remain profitable at 0.05%, confirming economic viability under realistic cost assumptions. Complete results are in Tables B15 and B16 in the Appendix.

6.3 Parameter Stability

To test whether results depend on specific parameter choices, we conduct sensitivity across four core parameters: Z-score entry $\{2.5, 3.0, 3.5\}$, exit threshold $\{0.3, 0.5, 0.7\}$, lookback $\{45, 60, 90\}$ min, and position size $\{20\%, 30\%, 40\%\}$. When additional data are available, we also include model parameters: centrality weight ω_f from Equation 11, the κ parameters BASE, α_{edge} , and α_{start} from

Equation 12, and slippage weight w_1 from Equation 14. This approach helps mitigate data snooping bias (White, 2000).

Figure 7 shows parameter sensitivity results. Strategies with mean Sharpe at or below zero are drawn in white; we attach limited interpretive value to parameter sensitivity for such strategies and do not emphasize them here. Although the entry threshold and lookback window can move some strategies from negative to positive Sharpe, we treat strategies whose sign is highly sensitive to parameters as having low stability and therefore do not focus on strategies with Sharpe below zero. Higher Z-score and longer lookback generally improve Sharpe; position-size effects are roughly linear; exit-threshold effects are strategy-specific. Successful strategies remain in positive-Sharpe regions across most parameter values.

The model parameters ω_f , κ , and w_1 shown in panels e–i exhibit flat sensitivity because contract markets have sparse connectivity: each currency has fewer tradable pairs than in spot markets, limiting the set of candidate paths. As a result, varying these parameters often leaves the chosen path unchanged. For the centrality parameters (ω_f) and κ parameters, we separately ran the S_2 path optimization routine (Algorithm 2) and confirmed that no path changes occurred across different parameter values. To maintain consistency, we therefore present sensitivity results using Sharpe ratio as the evaluation metric. This flat sensitivity confirms that our parameter choices do not strongly influence the optimal path selection, consistent with the limited connectivity of the contract market. Complete results for all 64 strategies are in Tables B6, B8, and B9 in the Appendix.

6.4 Walk-Forward Analysis

We conduct a walk-forward analysis over rolling windows to assess temporal stability. We divide the full sample period into multiple overlapping windows, compute Sharpe ratios for each window, and analyze the distribution of window-level performance metrics. This approach tests whether strategy performance is consistent across different time periods or driven by specific market conditions.

Table B4 (see Appendix) reports the average window Sharpe ratio and the coefficient of variation for each strategy. Most strategies with positive full-sample Sharpe maintain positive average window Sharpe and moderate coefficient of variation values, indicating that performance is not driven by a single sub-period. Strategies with low coefficient of variation values show stable performance across time windows, while higher coefficient of variation values suggest greater temporal variation in returns.

6.5 Out-of-Sample Testing

We split the sample into a 70% training period and a 30% testing period to evaluate generalization. The training period is used to estimate strategy parameters and calibrate the model, while the testing period evaluates performance on unseen data. This temporal split helps identify whether strategies are overfitted to specific sample characteristics or generalize to out-of-sample periods.

Table B5 (see Appendix) reports train/test Sharpe ratios and the degradation metric, defined as test Sharpe divided by train Sharpe. Of the 64 strategies, 63 (98%) exhibit degradation above 0.7, indicating low overfitting risk and strong generalization beyond the training period. Only one strategy, DIA (0.664), shows degradation below 0.7, indicating potential overfitting to the training period. Several high-performing strategies demonstrate exceptional generalization: CTK (1.066), IMX (1.007), SFP (1.043), ADX (1.039), and MANA (1.033) achieve degradation values exceeding 1.0, indicating that test-period performance actually exceeds training-period performance. These results validate that nearly all strategies generalize well to out-of-sample data, confirming minimal overfitting and supporting the robustness of our findings.

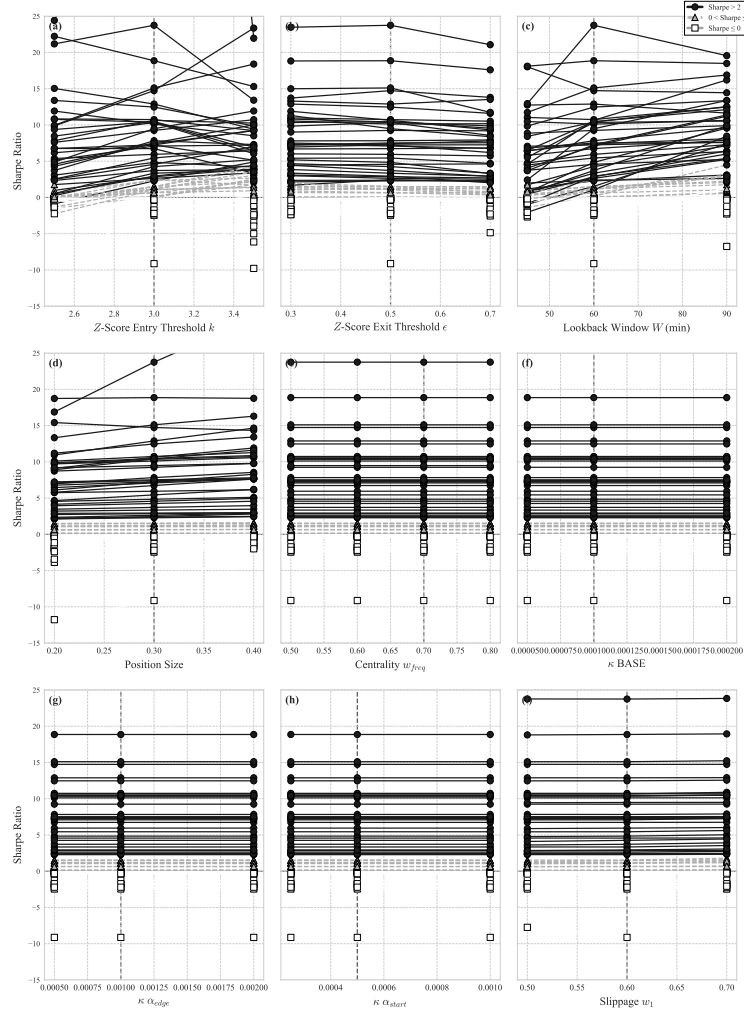


Figure 7: Parameter Stability. Each panel shows one parameter, with all 64 strategies as lines and points shaded by Sharpe; legend, upper right. Vertical dashed lines = baseline. Panels a–d: Z-score entry, exit threshold, lookback, position size; panels e–i show w_{freq} , κ BASE, $\kappa \alpha_{edge}$, $\kappa \alpha_{start}$, and w_1 when extra data are available. Lines in white: mean Sharpe ≤ 0 .

7 Conclusion

This paper draws on the arbitrage logic of Modigliani and Miller (1958) to establish a theoretical framework for multi-path arbitrage where different trading paths represent analogous “capital structures” for acquiring the same asset. We define V_1 and V_2 as the equivalent path-implied prices along the direct and indirect paths, respectively, and prove the no-arbitrage equilibrium condition $V_1 = V_2$, generalizing it to $k + 1$ trading pairs. When this equilibrium condition is violated due to transaction frictions and market microstructure imperfections, arbitrage opportunities emerge. The HJB value function is discretized into a DP path search, with each DP table entry directly implementing the Bellman recursion. While the theoretical framework applies to arbitrage paths with any number of intermediary assets, our path optimization algorithm systematically explores paths with 2 to k trading pairs, and empirically finds that triangular arbitrage paths are consistently

selected as optimal across all groups.

Our empirical analysis using high-frequency Binance data from January to December 2025 across 64 cryptocurrency trading pair groups validates three empirical predictions. First, we confirm prediction P1: arbitrage opportunities can arise in cryptocurrency contract markets, and some triangular-arbitrage opportunities remain economically exploitable after transaction costs; however, net profitability varies substantially across asset groups due to differences in liquidity, execution frictions, and market structure.

Second, we validate prediction P2: market network structure significantly affects arbitrage profitability. Strategies involving hub currencies are more likely to generate losses due to efficient price discovery, while those involving peripheral currencies show greater profitability potential.

Third, we validate prediction P3: network-based path optimization effectively improves arbitrage performance when applied to non-hub currencies with multiple viable paths. Strategy S_1 generated positive returns for 64.06% of all groups, while optimized strategy S_2 elevated this proportion to 75.00%.

Extensive robustness analysis confirms the reliability and practical applicability of our findings. Through five comprehensive tests including bootstrap resampling for statistical significance, transaction cost sensitivity for economic viability, parameter stability for methodological robustness, walk-forward analysis for temporal stability, and out-of-sample testing for generalization, we establish that our results represent genuine arbitrage opportunities rather than measurement artifacts.

This paper makes three contributions. First, drawing on the arbitrage logic of Modigliani and Miller (1958), we formalize trading routes as alternative capital structures, prove the equilibrium condition $V_1 = V_2$, and generalize it to $k + 1$ trading pairs. Second, we develop a network-based path-optimization approach for multi-path arbitrage under execution frictions without requiring price prediction, combining an HJB formulation of execution costs with discrete DP on the trading network, implemented as a Bellman recursion. Third, using high-frequency Binance data from January to December 2025 across 64 trading-pair groups, we present empirical results and robustness tests showing that the optimized strategy outperforms a baseline strategy in risk-adjusted performance.

We acknowledge a methodological limitation: for each currency, we evaluated only two paths, one under S_1 and one under S_2 . While potentially superior paths may exist for individual currencies, identifying them would require exhaustive backtesting of all feasible paths, which diverges from our objective of network-driven path selection. Future research directions include developing methods to identify optimal paths beyond exhaustive backtesting, such as advanced graph algorithms or machine learning approaches that can leverage network structure, as well as exploring more sophisticated network-based models that can better capture the dynamic relationships between network structure and arbitrage profitability.

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A Performance Metrics Definitions

This section provides formal definitions of the performance metrics used throughout the paper. All metrics are calculated based on daily returns $\{r_d\}_{d=1}^D$, where r_d represents the daily return on day d and D is the total number of trading days. For cryptocurrency markets, we use 365 trading days per year to reflect continuous 24/7 trading.

A.1 Return and Volatility Metrics

Annualized Return:

$$R_{\text{annual}} = \left[\left(\prod_{d=1}^D (1 + r_d) \right)^{365/D} - 1 \right] \times 100\%$$

where D is the actual number of trading days in the backtest period. The formula uses compound annualization, accounting for the compounding effect of daily returns. For cryptocurrency markets, we use 365 days per year instead of the traditional 252 trading days.

Annualized Volatility:

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{365} \times 100\%$$

where $\sigma_{\text{daily}} = \sqrt{\frac{1}{D-1} \sum_{d=1}^D (r_d - \bar{r})^2}$ is the unbiased daily standard deviation of returns (using $D - 1$ degrees of freedom), and $\bar{r} = \frac{1}{D} \sum_{d=1}^D r_d$ is the mean daily return.

A.2 Risk-Adjusted Return Metrics

Sharpe Ratio:

$$\text{Sharpe} = \frac{R_{\text{annual}}}{\sigma_{\text{annual}}}$$

where R_{annual} is the annualized return (in percentage) and σ_{annual} is the annualized volatility (in percentage). The risk-free rate is assumed to be zero for cryptocurrency markets. The Sharpe ratio measures risk-adjusted return per unit of total volatility.

Sortino Ratio:

$$\text{Sortino} = \frac{\bar{r}}{\sigma_{\text{down}}} \times \sqrt{365}$$

where $\bar{r}$ is the mean daily return, and σ_{down} is the standard deviation of downside returns (negative returns only). Specifically, $\sigma_{\text{down}} = \sqrt{\frac{1}{N_{\text{down}}} \sum_{r_d < 0} r_d^2}$ where N_{down} is the number of negative daily returns. The Sortino ratio measures excess return per unit of downside risk, focusing only on negative returns.

Calmar Ratio:

$$\text{Calmar} = \frac{R_{\text{annual}}}{\text{MaxDD}}$$

where R_{annual} is the annualized return (in percentage) and MaxDD is the maximum drawdown (in percentage, defined below). The Calmar ratio measures annualized return relative to the maximum drawdown, providing a risk-adjusted performance metric that emphasizes tail risk.

A.3 Drawdown Metrics

Maximum Drawdown (MaxDD):

$$\text{MaxDD} = \max_{1 \leq t \leq T} \left[\max_{1 \leq s \leq t} C_s - C_t \right] \times 100\%$$

where C_t is the cumulative return at time t , calculated as $C_t = \prod_{s=1}^t (1 + r_s) - 1$ with $C_0 = 0$. Maximum drawdown represents the largest peak-to-trough decline in cumulative returns over the entire period, expressed as a percentage.

A.4 Trading Performance Metrics

Win Rate:

$$\text{WinRate} = \frac{N_{\text{win}}}{N_{\text{total}}} \times 100\%$$

where N_{win} is the number of profitable trades (trades with positive return) and N_{total} is the total number of trades executed.

Cost Ratio:

$$\text{CostRatio} = \frac{\text{Total Transaction Costs}}{\text{Total Return}} \times 100\%$$

where total transaction costs (in percentage of initial capital) is calculated as $\text{Total Costs} = N_{\text{total}} \times 3 \times c \times 2 \times p \times 100\%$, with N_{total} being the total number of trades, c the transaction cost per leg (0.05% baseline), and p the position size (e.g., 30%). The factor of 3 accounts for the three legs in triangular arbitrage, and the factor of 2 accounts for both entry and exit. Total return is the cumulative return over the backtest period (in percentage). A negative CostRatio indicates that transaction costs exceed gross returns.

B Additional Tables

Table B1: S_1 Performance Statistics - Complete Results

Strategy	Return (%)	Sharpe	Sortino	Calmar	MaxDD (%)	Vol (%)	WinRate (%)	CostRatio (%)
01.XRP→ETH→USDT	-100.00%	-0.27	-0.10	-0.34	298.80%	294.89%	7.29%	-132.56%
02.INJ→BTC→USDT	24.47%	1.82	10.03	3.64	6.71%	5.21%	93.97%	1179.97%
03.DOT→BNB→USDT	2.91%	0.32	0.98	0.23	12.76%	5.38%	79.03%	5830.63%
04.SUI→EUR→USDT	33.20%	1.21	8.14	1.34	24.76%	7.04%	87.17%	1207.19%
05.LINK→ETH→USDT	-100.00%	-0.34	-0.23	-0.79	126.23%	300.61%	46.20%	-205.50%
06.UNI→BTC→USDT	-26.91%	-2.45	-13.69	-0.99	27.29%	4.17%	88.25%	-898.08%
07.LISTA→BNB→USDT	215.10%	10.64 [†]	32.22	29.58	7.27%	15.81%	71.59%	132.40%
08.LRC→ETH→USDT	25.77%	1.54	5.78	2.65	9.73%	10.57%	72.66%	969.23%
09.IOTA→BTC→USDT	-100.00%	-0.18	-0.55	-0.31	325.29%	494.76%	36.81%	-92.10%
10.MANTA→BTC→USDT	94.33%	8.06	28.78	6.37	14.81%	4.95%	88.38%	146.70%
11.VET→BNB→USDT	-21.88%	-0.92	-3.91	-0.65	33.71%	10.50%	69.19%	-1029.52%
12.GALA→USDC→USDT	-100.00%	-0.19	-0.88	-0.68	147.19%	556.28%	61.66%	-321.28%
13.MOVE→BTC→USDT	117.83%	4.93	27.65	16.98	6.94%	7.49%	78.35%	210.36%
14.POL→EUR→USDT	138.00%	14.74 [†]	71.90	118.01 [†]	1.17%	6.60%	95.00%	178.75%
15.PNUT→BTC→USDT	-100.00%	-0.29	1.68	-0.16	638.99%	571.80%	42.96%	-36.77%
16.AGLD→BTC→USDT	132.55%	5.95	70.83	54.06	2.45%	5.67%	92.89%	168.65%
17.REZ→USDC→USDT	259.99%	9.25	215.64	42.54	6.11%	7.21%	98.84%	141.55%
18.XLM→ETH→USDT	-100.00%	-0.18	2.33	-0.43	234.67%	551.21%	20.47%	-148.96%
19.UMA→BTC→USDT	69.57%	2.45	1.54	2.19	31.70%	43.39%	63.68%	423.80%
20.TRX→ETH→USDT	-100.00%	-0.33	-1.31	-0.93	107.37%	300.62%	53.38%	-243.26%
21.ARKM→FDUSD→USDT	283.90%	11.22 [†]	152.82	6.35	44.71%	6.78%	94.75%	77.25%
22.STEEM→ETH→USDT	42.83%	3.34	16.54	5.27	8.13%	6.22%	86.66%	454.33%
23.CELR→BTC→USDT	4812.64%	51.89 [†]	59.97	725.50 [†]	6.63%	34.98%	59.52%	43.65%
24.WIF→EUR→USDT	324.72%	10.12 [†]	85.48	50.91	6.38%	10.02%	94.60%	65.11%
25.ADX→ETH→USDT	1.89%	0.09	1.10	0.10	19.23%	7.74%	89.02%	11539.24%
26.1000SATS→FDUSD→USDT	151.68%	6.41	14.45	6.28	24.15%	9.29%	84.81%	205.41%
27.FET→BNB→USDT	127.73%	5.44	37.40	2.64	48.35%	14.93%	94.07%	186.06%
28.ETH→BTC→USDT	-24.34%	-8.19	-13.98	-1.06	23.06%	2.30%	1.16%	-101.09%
29.SOL→TUSD→USDT	37.62%	2.85	13.22	7.04	5.35%	4.35%	84.65%	532.82%
30.MEME→USDC→USDT	156.81%	4.57	33.92	6.61	23.74%	8.13%	90.40%	256.55%
31.TFUEL→BTC→USDT	326.29%	15.11 [†]	40.27	46.38	7.04%	12.08%	69.06%	106.80%
32.WIN→EUR→USDT	-100.00%	-0.06	9.55	-0.42	236.29%	1826.21%	2.31%	-94.12%
33.CTK→BNB→USDT	98.62%	7.09	14.38	9.53	10.35%	10.01%	80.38%	252.89%
34.ENA→BTC→USDT	117.05%	8.18	87.88	69.13	1.69%	4.10%	98.24%	209.76%
35.PYTH→BTC→USDT	66.64%	6.22	19.23	12.02	5.54%	5.01%	79.92%	328.04%
36.AI→BTC→USDT	156.33%	6.60	32.84	56.49	2.77%	13.95%	83.30%	139.99%
37.DASH→ETH→USDT	88.95%	1.12	11.24	2.23	39.85%	21.29%	69.20%	302.28%
38.HBAR→BTC→USDT	34.72%	3.97	48.10	72.86	0.48%	2.35%	99.14%	130.05%
39.CHZ→BNB→USDT	-57.74%	-1.26	-7.76	-0.80	72.55%	35.31%	57.51%	-417.74%
40.ONT→USDC→USDT	139.12%	7.41	61.00	51.07	2.72%	7.87%	94.42%	181.94%
41.CATI→FDUSD→USDT	56.94%	1.93	2.94	7.33	7.76%	29.44%	56.25%	-554.33%
42.BAT→BTC→USDT	51.50%	3.72	13.20	10.41	4.94%	5.60%	77.09%	466.89%
43.GRT→ETH→USDT	-27.26%	-1.51	-5.45	-0.90	30.46%	6.42%	74.02%	-783.55%
44.BOME→FDUSD→USDT	146.20%	4.63	13.30	6.79	21.54%	9.21%	86.70%	206.00%
45.NEIRO→USDC→USDT	-100.00%	-0.16	-1.21	-0.60	167.64%	650.43%	55.96%	-462.31%
46.RARE→USDC→USDT	169.29%	7.25	97.94	10.73	15.77%	6.61%	98.03%	142.27%
47.ICP→EUR→USDT	142.45%	7.25	49.93	48.07	2.96%	9.68%	87.10%	142.27%
48.IMX→BTC→USDT	173.97%	18.86 [†]	91.16	81.57	2.13%	5.10%	96.45%	121.52%
49.RONIN→FDUSD→USDT	-100.00%	-0.09	-0.58	-0.91	109.69%	2143.42%	50.84%	-234.17%
50.SUSHI→BTC→USDT	31.32%	1.49	8.33	1.78	17.56%	6.46%	89.64%	870.05%
51.DIA→BTC→USDT	48.16%	4.25	26.37	23.86	2.02%	4.70%	91.70%	403.58%
52.BNB→TUSD→USDT	-73.38%	-2.26	-16.08	-1.03	71.29%	10.23%	56.53%	-296.95%
53.YGG→USDC→USDT	142.83%	7.52	61.25	21.65	6.60%	5.29%	95.90%	193.67%
54.SFP→BTC→USDT	-62.45%	-1.72	-10.54	-1.03	60.33%	15.01%	58.29%	-406.04%
55.ZRX→BTC→USDT	33.09%	2.94	12.62	7.88	4.20%	5.46%	82.24%	617.28%
56.LTC→BTC→USDT	-84.00%	-2.71	-24.65	-1.02	82.26%	7.49%	45.13%	-202.43%
57.FLOKI→USDC→USDT	-100.00%	-0.07	-0.66	-0.37	271.36%	1599.72%	41.86%	-202.49%
58.BTC→FDUSD→USDT	-43.11%	-7.88	-20.17	-1.05	41.14%	3.18%	0.76%	-86.41%
59.AAVE→BTC→USDT	-100.00%	-0.39	-0.02	-0.74	135.08%	292.71%	64.74%	-268.87%
60.MANA→ETH→USDT	-35.31%	-2.21	-7.97	-1.04	33.95%	10.26%	66.74%	-710.14%
61.DOGE→BTC→USDT	39.12%	5.90	22.98	44.28	0.88%	2.46%	89.28%	335.53%
62.AVAX→EUR→USDT	77.64%	3.81	29.96	4.44	17.50%	18.85%	92.37%	387.10%
63.EOS→ETH→USDT	7.10%	1.05	2.34	1.33	5.35%	4.63%	84.01%	3252.72%
64.SXP→BTC→USDT	-100.00%	-0.35	-0.04	-0.43	231.04%	496.53%	47.71%	-114.72%

Complete performance statistics for all 64 groups using the baseline triangular arbitrage strategy S_1 . Groups with negative returns highlight the importance of careful asset selection in arbitrage strategies. MaxDD = Maximum Drawdown, Vol = Annual Volatility, WinRate = Win Rate, CostRatio = Cost Ratio. [†] indicates extreme values (Sharpe > 10 or Calmar > 100) that should be interpreted with caution, as these metrics may fail in extreme cases when volatility or drawdown approaches zero, which is a limitation of the metrics themselves rather than a calculation error. Note that Sortino ratio may also exhibit extreme values in similar cases, but these are not marked.

Table B2: S_2 Performance Statistics - Complete Results

Strategy	S_2 Ret (%)	Sharpe	Sortino	Calmar	MaxDD (%)	Vol (%)	WinRate (%)	S_1 Ret (%)	Status
01.XRP→TUSD→USDT	1.93	0.17	0.55	0.17	11.75	6.39	77.32	-100.00	Optimized
02.INJ→BNB→USDT	64.81	4.84	14.96	8.48	7.64	7.34	79.05	24.47	Optimized
03.DOT→EUR→USDT	-0.95	-0.04	0.84	-0.02	48.11	29.75	85.32	2.91	Failed
04.SUI→EUR→USDT	33.20	1.21	8.14	1.34	24.76	7.04	87.17	33.20	Optimized
05.LINK→EUR→USDT	28.40	1.49	12.22	1.25	22.77	15.33	92.21	-100.00	Optimized
06.UNI→ETH→USDT	-89.49	-9.13	-16.70	-0.95	94.09	30.12	59.30	-26.91	Failed
07.LISTA→BNB→USDT	215.10	10.64 [†]	32.22	29.58	7.27	15.81	71.59	215.10	Optimized
08.LRC→ETH→USDT	25.77	1.54	5.78	2.65	9.73	10.57	72.66	25.77	Optimized
09.IOTA→ETH→USDT	-56.52	-1.80	-10.37	-0.83	67.76	12.04	60.84	-100.00	Optimized
10.MANTA→USDC→USDT	113.76	6.73	27.47	22.14	5.14	6.34	95.24	94.33	Optimized
11.VET→EUR→USDT	144.19	7.83	80.90	105.67 [†]	1.36	15.73	96.22	-21.88	Optimized
12.GALA→EUR→USDT	265.92	10.74 [†]	50.22	52.47	5.07	16.43	86.71	-100.00	Optimized
13.MOVE→BNB→USDT	84.45	2.63	13.03	1.83	46.12	11.18	71.52	117.83	Failed
14.POL→EUR→USDT	138.00	14.74 [†]	71.90	118.01 [†]	1.17	6.60	95.00	138.00	Optimized
15.PNUT→EUR→USDT	965.28	12.88 [†]	40.12	24.60	39.24	23.46	52.63	-100.00	Optimized
16.AGLD→BTC→USDT	132.55	5.95	70.83	54.06	2.45	5.67	92.89	132.55	Optimized
17.REZ→USDC→USDT	259.99	9.25	215.64	42.54	6.11	7.21	98.84	259.99	Optimized
18.XLM→EUR→USDT	51.02	2.28	20.17	2.80	18.25	6.12	90.47	-100.00	Optimized
19.UMA→BTC→USDT	69.57	2.45	1.54	2.19	31.70	43.39	63.68	69.57	Optimized
20.TRX→EUR→USDT	-100.00	-0.32	-0.21	-0.63	159.01	318.56	23.68	-100.00	Optimized
21.ARKM→BNB→USDT	295.47	12.47 [†]	90.59	3.55	83.24	16.09	90.31	283.90	Optimized
22.STEEM→ETH→USDT	42.83	3.34	16.54	5.27	8.13	6.22	86.66	42.83	Optimized
23.CELR→BTC→USDT	4812.64	51.89 [†]	59.97	725.50 [†]	6.63	34.98	59.52	4812.64	Optimized
24.WIF→EUR→USDT	324.72	10.12 [†]	85.48	50.91	6.38	10.02	94.60	324.72	Optimized
25.ADX→ETH→USDT	1.89	0.09	1.10	0.10	19.23	7.74	89.02	1.89	Optimized
26.1000SATS→USDC→USDT	59.92	2.37	32.86	5.21	11.51	10.52	92.63	151.68	Failed
27.FET→BNB→USDT	127.73	5.44	37.40	2.64	48.35	14.93	94.07	127.73	Optimized
28.ETH→EUR→USDT	-47.95	-1.84	-15.34	-0.76	63.19	9.76	46.47	-24.34	Failed
29.SOL→TUSD→USDT	37.62	2.85	13.22	7.04	5.35	4.35	84.65	37.62	Optimized
30.MEME→USDC→USDT	156.81	4.57	33.92	6.61	23.74	8.13	90.40	156.81	Optimized
31.TFUEL→BTC→USDT	326.29	15.11 [†]	40.27	46.38	7.04	12.08	69.06	326.29	Optimized
32.WIN→EUR→USDT	-100.00	-0.06	9.55	-0.42	236.29	1826.21	2.31	-100.00	Optimized
33.CTK→BNB→USDT	98.62	7.09	14.38	9.53	10.35	10.01	80.38	98.62	Optimized
34.ENA→BNB→USDT	280.30	10.40 [†]	103.39	22.40	12.51	24.94	89.37	117.05	Optimized
35.PYTH→USDC→USDT	24.41	0.63	7.72	0.47	51.54	7.50	85.96	66.64	Failed
36.AI→BNB→USDT	170.41	10.35 [†]	32.31	13.96	12.21	11.04	81.65	156.33	Optimized
37.DASH→ETH→USDT	88.95	1.12	11.24	2.23	39.85	21.29	69.20	88.95	Optimized
38.HBAR→BNB→USDT	-100.00	-0.29	0.47	-0.47	212.50	355.42	35.22	34.72	Failed
39.CHZ→BNB→USDT	-57.74	-1.26	-7.76	-0.80	72.55	35.31	57.51	-57.74	Optimized
40.ONT→USDC→USDT	139.12	7.41	61.00	51.07	2.72	7.87	94.42	139.12	Optimized
41.CATI→BNB→USDT	-99.97	-2.46	-23.23	-3.54	28.22	40.73	45.95	56.94	Failed
42.BAT→BTC→USDT	51.50	3.72	13.20	10.41	4.94	5.60	77.09	51.50	Optimized
43.GRT→EUR→USDT	161.51	10.46 [†]	34.49	41.35	3.91	11.13	86.85	-27.26	Optimized
44.BOME→EUR→USDT	704.23	23.75 [†]	51.56	45.46	15.49	20.61	81.51	146.20	Optimized
45.NEIRO→EUR→USDT	2017.48	157.52 [†]	259.77	928.09 [†]	2.17	11.55	84.94	-100.00	Optimized
46.RARE→USDC→USDT	169.29	7.25	97.94	10.73	15.77	6.61	98.03	169.29	Optimized
47.ICP→EUR→USDT	142.45	7.25	49.93	48.07	2.96	9.68	87.10	142.45	Optimized
48.IMX→BTC→USDT	173.97	18.86 [†]	91.16	81.57	2.13	5.10	96.45	173.97	Optimized
49.ROBIN→FUDUSD→USDT	-100.00	-0.09	-0.58	-0.91	109.69	2143.42	50.84	-100.00	Optimized
50.SUSHI→BTC→USDT	31.32	1.49	8.33	1.78	17.56	6.46	89.64	31.32	Optimized
51.DIA→BTC→USDT	48.16	4.25	26.37	23.86	2.02	4.70	91.70	48.16	Optimized
52.BNB→TUSD→USDT	-73.38	-2.26	-16.08	-1.03	71.29	10.23	56.53	-73.38	Optimized
53.YGG→USDC→USDT	142.83	7.52	61.25	21.65	6.60	5.29	95.90	142.83	Optimized
54.SFP→BTC→USDT	-62.45	-1.72	-10.54	-1.03	60.33	15.01	58.29	-62.45	Optimized
55.ZRX→BTC→USDT	33.09	2.94	12.62	7.88	4.20	5.46	82.24	33.09	Optimized
56.LTC→EUR→USDT	-100.00	-0.17	-0.75	-0.51	197.81	461.40	49.30	-84.00	Failed
57.FLOKI→USDC→USDT	-100.00	-0.07	-0.66	-0.37	271.36	1599.72	41.86	-100.00	Optimized
58.BTC→EUR→USDT	-100.00	-0.28	0.34	-0.69	144.94	395.00	20.54	-43.11	Failed
59.AAVE→ETH→USDT	-100.00	-0.33	0.35	-0.75	133.71	432.54	46.34	-100.00	Optimized
60.MANA→ETH→USDT	-35.31	-2.21	-7.97	-1.04	33.95	10.26	66.74	-35.31	Optimized
61.DOGE→TUSD→USDT	88.51	9.48	16.30	14.71	6.02	8.38	74.34	39.12	Optimized
62.AVAX→BNB→USDT	14.41	1.14	5.77	0.88	16.44	4.47	88.09	77.64	Failed
63.EOS→ETH→USDT	7.10	1.05	2.34	1.33	5.35	4.63	84.01	7.10	Optimized
64.SXP→BTC→USDT	33.30	0.66	6.38	1.17	28.36	13.26	71.66	-100.00	Optimized

Complete results for all 64 groups. "Status" indicates: Optimized: S_2 path optimization improved negative returns to positive ($S_2 \geq S_1$); Failed: optimization could not achieve positive returns ($S_2 < S_1$). MaxDD = Maximum Drawdown, Vol = Annual Volatility, WinRate = Win Rate. [†] indicates extreme values (Sharpe > 10 or Calmar > 100) that should be interpreted with caution. For arbitrage strategies, Sharpe and Calmar may fail in extreme cases when volatility or drawdown approaches zero, which is a limitation of the metrics themselves rather than a calculation error. Note that Sortino ratio may also exhibit extreme values in similar cases, but these are not marked.

Table B3: Bootstrap Statistical Validation - Complete Results

Strategy	Obs Sharpe	Bootstrap Mean	Std Dev	CI Lower	CI Upper	P-Value	Sig
01_XRP→TUSD→USDT	0.174	0.266	1.021	-1.751	2.134	0.396	No
02_INJ→BNB→USDT	4.841	7.060	1.039	5.336	9.312	0.000	Yes
03_DOT→EUR→USDT	-0.039	-1.426	2.498	-6.075	1.452	0.516	No
04_SUI→EUR→USDT	1.208	4.115	0.923	2.355	5.983	0.000	Yes
05_LINK→EUR→USDT	1.486	2.020	0.681	0.904	3.739	0.003	Yes
06_UNI→ETH→USDT	-9.134	-9.881	5.029	-18.377	-2.885	1.000	No
07_LISTA→BNB→USDT	10.639	7.499	0.749	6.100	8.985	0.000	Yes
08_LRC→ETH→USDT	1.537	2.135	0.908	0.209	3.870	0.011	Yes
09_IOTA→ETH→USDT	-1.805	-6.949	1.267	-9.481	-4.485	1.000	No
10_MANTA→USDC→USDT	6.733	12.779	2.195	9.284	16.831	0.000	Yes
11_VET→EUR→USDT	7.827	8.649	4.215	4.425	15.071	0.000	Yes
12_GALA→EUR→USDT	10.741	9.189	2.466	6.100	14.630	0.000	Yes
13_MOVE→BNB→USDT	2.633	5.594	0.893	3.869	7.433	0.000	Yes
14_POL→EUR→USDT	14.736	13.966	2.304	10.094	17.794	0.000	Yes
15_PNUT→EUR→USDT	12.884	10.318	0.712	8.948	11.773	0.000	Yes
16_AGLD→BTC→USDT	5.946	14.972	0.723	13.557	16.488	0.000	Yes
17_REZ→USDC→USDT	9.247	18.018	1.064	16.077	20.119	0.000	Yes
18_XLM→EUR→USDT	2.283	6.780	0.868	5.078	8.542	0.000	Yes
19_UMA→BTC→USDT	2.451	1.508	1.131	-0.662	3.746	0.095	No
20_TRX→EUR→USDT	-0.321	-0.300	1.030	-2.213	1.518	0.599	No
21_ARKM→BNB→USDT	12.468	9.835	2.793	6.770	17.421	0.000	Yes
22_STEEM→ETH→USDT	3.343	5.842	0.748	4.480	7.427	0.000	Yes
23_CELR→BTC→USDT	51.890	11.478	1.375	8.939	14.264	0.000	Yes
24_WIF→EUR→USDT	10.117	14.894	1.515	12.342	18.380	0.000	Yes
25_ADX→ETH→USDT	0.088	0.058	1.095	-2.413	1.901	0.441	No
26_1000SATS→USDC→USDT	2.368	5.199	1.241	3.558	7.592	0.000	Yes
27_FET→BNB→USDT	5.442	8.148	3.818	4.363	15.106	0.000	Yes
28_ETH→EUR→USDT	-1.841	-7.058	2.038	-11.481	-3.503	1.000	No
29_SOL→TUSD→USDT	2.852	7.377	0.948	5.468	9.195	0.000	Yes
30_MEME→USDC→USDT	4.569	11.723	0.844	10.167	13.495	0.000	Yes
31_TFUEL→BTC→USDT	15.107	12.117	0.834	10.518	13.718	0.000	Yes
32_WIN→EUR→USDT	-0.057	0.926	0.831	-1.375	2.012	0.137	No
33_CTK→BNB→USDT	7.089	6.950	0.824	5.378	8.589	0.000	Yes
34_ENA→BNB→USDT	10.403	9.761	5.927	4.292	18.506	0.000	Yes
35_PYTH→USDC→USDT	0.629	2.927	0.953	1.067	4.792	0.003	Yes
36_AI→BNB→USDT	10.348	9.779	1.760	6.988	12.952	0.000	Yes
37_DASH→ETH→USDT	1.125	3.049	0.822	1.440	4.517	0.001	Yes
38_HBAR→BNB→USDT	-0.292	0.169	1.012	-1.942	1.852	0.400	No
39_CHZ→BNB→USDT	-1.261	-3.702	2.878	-9.704	0.190	0.965	No
40_ONT→USDC→USDT	7.410	11.405	1.010	9.701	13.639	0.000	Yes
41_CATI→BNB→USDT	-2.455	-20.738	6.125	-34.680	-11.063	1.000	No
42_BAT→BTC→USDT	3.717	7.460	1.061	5.340	9.501	0.000	Yes
43_GRT→EUR→USDT	10.456	10.444	3.210	6.360	15.982	0.000	Yes
44_BOME→EUR→USDT	23.752	10.601	1.228	8.684	13.356	0.000	Yes
45_NEIRO→EUR→USDT	157.522	27.140	3.063	21.889	34.377	0.000	Yes
46_RARE→USDC→USDT	7.250	15.298	1.127	13.305	17.738	0.000	Yes
47_ICP→EUR→USDT	7.254	9.717	1.503	7.259	12.765	0.000	Yes
48_IMX→BTC→USDT	18.862	19.965	0.993	18.165	22.055	0.000	Yes
49_RONIN→FDUSD→USDT	-0.088	-0.305	1.100	-1.849	1.966	0.677	No
50_SUSHI→BTC→USDT	1.488	4.326	0.906	2.651	6.208	0.000	Yes
51_DIA→BTC→USDT	4.255	8.434	0.683	7.085	9.719	0.000	Yes
52_BNB→TUSD→USDT	-2.259	-13.039	1.556	-15.889	-9.835	1.000	No
53_YGG→USDC→USDT	7.522	16.926	1.010	14.944	18.927	0.000	Yes
54_SFP→BTC→USDT	-1.721	-6.622	1.437	-9.602	-3.910	1.000	No
55_ZRX→BTC→USDT	2.935	5.423	0.881	3.762	7.202	0.000	Yes
56_LTC→EUR→USDT	-0.170	-0.766	0.912	-2.022	1.508	0.819	No
57_FLOKI→USDC→USDT	-0.067	-0.617	1.036	-1.872	1.903	0.754	No
58_BTC→EUR→USDT	-0.285	-0.057	1.088	-2.322	1.678	0.458	No
59_AAVE→ETH→USDT	-0.332	-0.416	1.357	-2.955	1.559	0.508	No
60_MANA→ETH→USDT	-2.205	-4.613	1.731	-7.966	-1.537	1.000	No
61_DOGE→TUSD→USDT	9.482	7.676	0.913	5.929	9.461	0.000	Yes
62_AVAX→BNB→USDT	1.136	3.086	0.887	1.292	4.840	0.000	Yes
63_EOS→ETH→USDT	1.051	1.433	1.683	-1.897	4.717	0.208	No
64_SXP→BTC→USDT	0.659	2.146	0.883	0.310	3.845	0.014	Yes

Complete bootstrap resampling results (1000 iterations) for all 64 strategies. P-value tests the null hypothesis that Sharpe ratio equals zero. Strategies with $p < 0.05$ are considered statistically significant. 95% confidence intervals (CI Lower, CI Upper) provide range estimates for the true Sharpe ratio. High-performing peripheral currency strategies show strong statistical significance ($p < 0.001$), while hub currency strategies (ETH, BTC, LTC, BNB) show significant negative returns ($p = 1.000$).

Table B4: Walk-Forward Analysis - Summary Statistics

Strategy	Full Sharpe	Avg Window Sharpe	Std Dev	Min Sharpe	Max Sharpe	Coefficient of Variation
01.XRP→TUSD→USDT	0.174	-1.108	0.611	-2.005	-0.288	0.552
02.INJ→BNB→USDT	4.841	8.758	2.501	6.661	13.774	0.286
03.DOT→EUR→USDT	-0.039	-0.616	0.684	-1.509	0.469	1.111
04.SUI→EUR→USDT	1.208	3.586	0.571	2.773	4.535	0.159
05.LINK→EUR→USDT	1.486	1.548	0.862	0.127	2.510	0.557
06.UNI→ETH→USDT	-9.134	-3.096	0.599	-4.307	-2.605	0.193
07.LISTA→BNB→USDT	10.639	16.818	4.358	9.681	21.555	0.259
08.LRC→ETH→USDT	1.537	1.290	1.187	0.236	3.354	0.920
09.IOTA→ETH→USDT	-1.805	-6.788	3.550	-13.350	-4.165	0.523
10.MANTA→USDC→USDT	6.733	24.091	5.262	16.380	29.946	0.218
11.VET→EUR→USDT	7.827	13.391	5.355	9.197	25.143	0.400
12.GALA→EUR→USDT	10.741	15.754	4.264	12.024	23.976	0.271
13.MOVE→BNB→USDT	2.633	6.456	1.709	4.161	9.716	0.265
14.POL→EUR→USDT	14.736	31.941	4.209	27.980	39.589	0.132
15.PNUT→EUR→USDT	12.884	34.610	8.118	23.112	46.044	0.235
16.AGLD→BTC→USDT	5.946	26.935	4.807	18.583	32.668	0.178
17.REZ→USDC→USDT	9.247	48.874	11.847	34.238	62.202	0.242
18.XLM→EUR→USDT	2.283	8.473	3.409	5.438	14.860	0.402
19.UMA→BTC→USDT	2.451	0.786	0.847	-0.175	2.060	1.078
20.TRX→EUR→USDT	-0.321	9.779	9.098	-0.191	19.373	0.930
21.ARKM→BNB→USDT	12.468	13.964	2.000	11.475	16.945	0.143
22.STEEM→ETH→USDT	3.343	10.311	1.494	8.022	12.456	0.145
23.CELR→BTC→USDT	51.890	157.995	57.673	92.783	257.550	0.365
24.WIF→EUR→USDT	10.117	23.654	5.784	19.156	35.007	0.245
25.ADX→ETH→USDT	0.088	2.338	2.689	-1.392	6.145	1.150
26.1000SATS→USDC→USDT	2.368	3.848	1.045	2.579	5.278	0.272
27.FET→BNB→USDT	5.442	8.685	3.108	5.252	13.123	0.358
28.ETH→EURI→USDT	-1.841	-7.942	1.830	-9.947	-5.277	0.230
29.SOL→TUSD→USDT	2.852	6.578	2.638	3.596	9.907	0.401
30.MEME→USDC→USDT	4.569	16.722	4.754	10.726	21.250	0.284
31.TFUEL→BTC→USDT	15.107	18.450	1.705	16.413	21.224	0.092
32.WIN→EUR→USDT	-0.057	295.778	461.085	1.445	969.846	1.559
33.CTK→BNB→USDT	7.089	7.252	1.489	5.554	9.602	0.205
34.ENA→BNB→USDT	10.403	22.752	2.585	19.679	26.019	0.114
35.PYTH→USDC→USDT	0.629	5.051	1.570	3.222	7.472	0.311
36.AI→BNB→USDT	10.348	17.934	3.722	13.166	24.953	0.208
37.DASH→ETH→USDT	1.125	1.921	1.852	-0.172	5.419	0.964
38.HBAR→BNB→USDT	-0.292	1.057	1.481	-0.724	3.337	1.400
39.CHZ→BNB→USDT	-1.261	-3.054	1.742	-5.863	-1.330	0.571
40.ONT→USDC→USDT	7.410	29.135	6.550	21.466	37.426	0.225
41.CATI→BNB→USDT	-2.455	-3.910	0.538	-4.151	-2.695	0.137
42.BAT→BTC→USDT	3.717	12.984	2.021	10.993	16.936	0.156
43.GRT→EUR→USDT	10.456	10.188	3.655	5.635	16.260	0.359
44.BOME→EUR→USDT	23.752	35.174	10.541	23.191	51.533	0.300
45.NEIRO→EUR→USDT	157.522	183.846	64.611	117.451	304.624	0.351
46.RARE→USDC→USDT	7.250	31.169	5.099	22.979	38.976	0.164
47.ICP→EUR→USDT	7.254	33.580	10.095	23.331	44.353	0.301
48.IMX→BTC→USDT	18.862	48.585	15.553	30.262	78.061	0.320
49.RONIN→FDUSD→USDT	-0.088	760.128	1117.984	6.836	2393.038	1.471
50.SUSHI→BTC→USDT	1.488	6.646	1.557	4.568	8.713	0.234
51.DIA→BTC→USDT	4.255	11.652	1.177	9.499	13.087	0.101
52.BNB→TUSD→USDT	-2.259	-6.316	1.638	-8.939	-4.473	0.259
53.YGG→USDC→USDT	7.522	35.557	8.029	27.502	48.415	0.226
54.SFP→BTC→USDT	-1.721	-4.617	0.834	-5.781	-3.583	0.181
55.ZRX→BTC→USDT	2.935	4.403	1.168	3.329	6.789	0.265
56.LTC→EUR→USDT	-0.170	-0.767	0.394	-1.353	-0.197	0.514
57.FLOKI→USDC→USDT	-0.067	309.574	505.004	-0.431	1207.414	1.631
58.BTC→EURI→USDT	-0.285	22.218	32.221	-0.141	78.699	1.450
59.AAVE→ETH→USDT	-0.332	1.696	3.249	-0.826	7.262	1.915
60.MANA→ETH→USDT	-2.205	-3.867	2.520	-7.722	-0.709	0.652
61.DOGE→TUSD→USDT	9.482	19.987	4.422	14.396	25.920	0.221
62.AVAX→BNB→USDT	1.136	2.723	0.272	2.432	3.141	0.100
63.EOS→ETH→USDT	1.051	2.548	1.175	1.171	4.797	0.461
64.SXP→BTC→USDT	0.659	2.136	1.002	1.288	4.083	0.469

Walk-forward analysis summary statistics for all 64 strategies based on 7 rolling 6-month windows. “Avg Window Sharpe” is the average Sharpe ratio across all windows, “Std Dev” measures temporal variability, and “Coefficient of Variation” ($CV = Std/Mean$) indicates relative stability. Lower CV values indicate more stable performance across time periods. The majority of successful strategies (positive full Sharpe) maintain positive average window Sharpe ratios with moderate CV values, indicating temporal robustness. Complete window-by-window results (450 rows) are available upon request from the authors.

Table B5: Out-of-Sample Testing - Complete Results

Strategy	Full Sharpe	Train Sharpe	Test Sharpe	Degradation	Train Ret(%)	Test Ret(%)	Overfitting
01.XRP→TUSD→USDT	0.174	0.304	0.262	0.860	1.93	1.61	Low
02.INJ→BNB→USDT	4.841	8.627	7.873	0.913	64.94	57.88	Low
03.DOT→EUR→USDT	-0.039	-0.031	-0.025	0.812	-0.95	-0.74	Low
04.SUI→EUR→USDT	1.208	4.639	3.705	0.799	33.27	25.64	Low
05.LINK→EUR→USDT	1.486	1.804	1.640	0.909	28.46	24.41	Low
06.UNI→ETH→USDT	-9.134	-2.995	-2.231	0.745	-89.67	-66.51	Low
07.LISTA→BNB→USDT	10.639	14.189	10.926	0.770	215.52	168.01	Low
08.LRC→ETH→USDT	1.537	2.354	1.769	0.751	25.82	18.82	Low
09.IOTA→ETH→USDT	-1.805	-4.521	-3.906	0.864	-56.63	-47.24	Low
10.MANTA→USDC→USDT	6.733	17.824	17.267	0.969	113.98	105.73	Low
11.VET→EUR→USDT	7.827	9.546	8.107	0.849	144.47	129.98	Low
12.GALA→EUR→USDT	10.741	16.620	14.940	0.899	266.44	246.95	Low
13.MOVE→BNB→USDT	2.633	7.274	6.913	0.950	84.62	78.63	Low
14.POL→EUR→USDT	14.736	21.455	20.375	0.950	138.27	133.13	Low
15.PNUT→EUR→USDT	12.884	39.472	31.440	0.796	967.17	731.25	Low
16.AGLD→BTC→USDT	5.946	22.501	18.640	0.828	132.81	107.57	Low
17.REZ→USDC→USDT	9.247	36.597	33.522	0.916	260.50	238.02	Low
18.XLM→EUR→USDT	2.283	8.186	5.917	0.723	51.12	37.45	Low
19.UMA→BTC→USDT	2.451	1.587	1.343	0.846	69.71	56.05	Low
20.TRX→EUR→USDT	-0.321	-0.328	-0.267	0.816	-100.20	-84.85	Low
21.ARKM→BNB→USDT	12.468	17.639	13.698	0.777	296.05	211.78	Low
22.STEEM→ETH→USDT	3.343	6.899	6.464	0.937	42.91	41.50	Low
23.CELR→BTC→USDT	51.890	131.417	116.313	0.885	4822.08	4154.88	Low
24.WIF→EUR→USDT	10.117	32.854	28.241	0.860	325.36	272.44	Low
25.ADX→ETH→USDT	0.088	0.239	0.248	1.039	1.89	1.86	Low
26.1000SATS→USDC→USDT	2.368	5.654	4.706	0.832	60.04	48.37	Low
27.FET→BNB→USDT	5.442	8.445	6.388	0.756	127.98	99.40	Low
28.ETH→EUR→USDT	-1.841	-5.134	-4.958	0.966	-48.04	-47.25	Low
29.SOL→TUSD→USDT	2.852	8.368	7.107	0.849	37.69	31.29	Low
30.MEME→USDC→USDT	4.569	20.116	15.662	0.779	157.12	125.27	Low
31.TFUEL→BTC→USDT	15.107	26.618	24.443	0.918	326.93	294.32	Low
32.WIN→EUR→USDT	-0.057	-0.055	-0.054	0.967	-100.20	-96.84	Low
33.CTK→BNB→USDT	7.089	9.465	10.086	1.066	98.81	97.15	Low
34.ENA→BNB→USDT	10.403	10.729	9.461	0.882	280.85	245.92	Low
35.PYTH→USDC→USDT	0.629	3.321	2.696	0.812	24.46	20.44	Low
36.AI→BNB→USDT	10.348	15.102	11.913	0.789	170.74	125.97	Low
37.DASH→ETH→USDT	1.125	4.355	3.607	0.828	89.12	79.88	Low
38.HBAR→BNB→USDT	-0.292	-0.283	-0.222	0.785	-100.20	-80.58	Low
39.CHZ→BNB→USDT	-1.261	-1.664	-1.600	0.962	-57.85	-56.09	Low
40.ONT→USDC→USDT	7.410	17.817	15.344	0.861	139.39	116.24	Low
41.CATI→BNB→USDT	-2.455	-2.521	-1.792	0.711	-100.17	-70.45	Low
42.BAT→BTC→USDT	3.717	9.159	8.863	0.968	51.60	48.48	Low
43.GRT→EUR→USDT	10.456	14.600	12.038	0.825	161.83	131.33	Low
44.BOME→EUR→USDT	23.752	34.612	27.892	0.806	705.61	550.08	Low
45.NEIRO→EUR→USDT	157.522	173.733	129.579	0.746	2021.44	1427.16	Low
46.RARE→USDC→USDT	7.250	26.218	21.125	0.806	169.62	140.88	Low
47.ICP→EUR→USDT	7.254	15.117	14.484	0.958	142.73	140.41	Low
48.IMX→BTC→USDT	18.862	33.682	33.928	1.007	174.31	167.11	Low
49.RONIN→FDUSD→USDT	-0.088	-0.046	-0.043	0.936	-100.20	-90.14	Low
50.SUSHI→BTC→USDT	1.488	4.975	4.793	0.963	31.38	29.70	Low
51.DIA→BTC→USDT	4.255	10.404	6.907	0.664	48.25	33.99	High
52.BNB→TUSD→USDT	-2.259	-7.536	-6.274	0.833	-73.52	-65.11	Low
53.YGG→USDC→USDT	7.522	25.769	19.682	0.764	143.11	106.59	Low
54.SFP→BTC→USDT	-1.721	-3.996	-4.167	1.043	-62.57	-61.09	Low
55.ZRX→BTC→USDT	2.935	6.242	5.269	0.844	33.15	27.46	Low
56.LTC→EUR→USDT	-0.170	-0.226	-0.166	0.734	-100.20	-78.39	Low
57.FLOKI→USDC→USDT	-0.067	-0.060	-0.055	0.919	-100.20	-84.96	Low
58.BTC→EUR→USDT	-0.285	-0.243	-0.243	0.999	-100.20	-94.80	Low
59.AAVE→ETH→USDT	-0.332	-0.224	-0.178	0.795	-100.20	-78.02	Low
60.MANA→ETH→USDT	-2.205	-3.305	-3.415	1.033	-35.38	-33.57	Low
61.DOGE→TUSD→USDT	9.482	10.944	9.208	0.841	88.68	74.14	Low
62.AVAX→BNB→USDT	1.136	3.134	2.583	0.824	14.44	11.72	Low
63.EOS→ETH→USDT	1.051	1.490	1.195	0.802	7.11	5.33	Low
64.SXP→BTC→USDT	0.659	2.419	1.883	0.779	33.37	24.98	Low

Out-of-sample testing results for all 64 strategies. Training period: first 70% (255 days); Testing period: last 30% (110 days). “Degradation” = Test Sharpe / Train Sharpe; values > 0.7 indicate low overfitting risk. Among 64 strategies, 63 show low overfitting risk (degradation > 0.7), demonstrating that most strategies generalize well beyond their training period. Only one strategy, DIA (degradation = 0.664), shows high overfitting risk, which should be interpreted with caution. Complete results validate temporal robustness of our findings.

Table B6: Parameter Sensitivity Analysis - Z-Entry Threshold Variations

Strategy	Z=2.5 Ret(%)	Z=2.5 Sharpe	Z=3.0 Ret(%)	Z=3.0 Sharpe	Z=3.5 Ret(%)	Z=3.5 Sharpe	Optimal Z
01.XRP→TUSD→USDT	-40.59	-1.49	1.93	0.17	6.27	1.76	3.5
02.INJ→BNB→USDT	80.80	3.07	64.81	4.84	31.19	4.21	3.0
03.DOT→EUR→USDT	-100.00	-0.14	-0.95	-0.04	45.07	2.47	3.5
04.SUI→EUR→USDT	-85.56	-0.70	33.20	1.21	34.43	3.04	3.5
05.LINK→EUR→USDT	-100.00	-0.10	28.40	1.49	48.78	3.95	3.5
06.UNI→ETH→USDT	-100.00	-0.06	-89.49	-9.13	-15.13	-3.28	N/A
07.LISTA→BNB→USDT	273.76	7.83	215.10	10.64	143.04	10.26	3.0
08.LRC→ETH→USDT	-34.04	-0.78	25.77	1.54	20.95	2.09	3.5
09.IOTA→ETH→USDT	-100.00	-0.21	-56.52	-1.80	-27.52	-2.59	N/A
10.MANTA→USDC→USDT	230.90	6.76	113.76	6.73	40.93	3.34	2.5
11.VET→EUR→USDT	123.48	3.72	144.19	7.83	95.74	7.30	3.0
12.GALA→EUR→USDT	383.72	10.83	265.92	10.74	163.01	9.22	2.5
13.MOVE→BNB→USDT	98.07	1.77	84.45	2.63	49.34	3.39	3.5
14.POL→EUR→USDT	161.08	9.87	138.00	14.74	81.99	23.36	3.5
15.PNUT→EUR→USDT	1378.87	15.04	965.28	12.88	570.30	9.41	2.5
16.AGLD→BTC→USDT	202.82	5.38	132.55	5.95	55.60	6.43	3.5
17.REZ→USDC→USDT	368.66	7.60	259.99	9.25	137.51	10.73	3.5
18.XLM→EUR→USDT	-52.18	-0.88	51.02	2.28	46.41	4.90	3.5
19.UMA→BTC→USDT	-25.96	-0.11	69.57	2.45	44.22	1.71	3.0
20.TRX→EUR→USDT	-100.00	-0.17	-100.00	-0.32	-46.81	-6.14	N/A
21.ARKM→BNB→USDT	395.56	13.38	295.47	12.47	153.55	9.87	2.5
22.STEEM→ETH→USDT	5.54	0.24	42.83	3.34	29.54	3.78	3.5
23.CELR→BTC→USDT	1970.28	24.42	4812.64	51.89	6799.80	80.55	3.5
24.WIF→EUR→USDT	647.59	11.92	324.72	10.12	140.09	8.50	2.5
25.ADX→ETH→USDT	-100.00	-0.28	1.89	0.09	23.67	2.56	3.5
26.1000SATS→USDC→USDT	33.22	0.65	59.92	2.37	44.26	2.67	3.5
27.FET→BNB→USDT	142.34	2.54	127.73	5.44	57.32	6.64	3.5
28.ETH→EUR→USDT	-100.00	-0.36	-47.95	-1.84	-0.52	-0.06	N/A
29.SOL→TUSD→USDT	25.87	0.97	37.62	2.85	17.04	3.93	3.5
30.MEME→USDC→USDT	159.90	2.49	156.81	4.57	90.86	7.02	3.5
31.TFUEL→BTC→USDT	309.10	9.95	326.29	15.11	309.88	18.40	3.5
32.WIN→EUR→USDT	-100.00	-0.13	-100.00	-0.06	-93.04	-1.44	N/A
33.CTK→BNB→USDT	100.96	5.14	98.62	7.09	68.46	7.23	3.5
34.ENA→BNB→USDT	445.70	10.77	280.30	10.40	126.01	6.50	2.5
35.PYTH→USDC→USDT	-100.00	-0.10	24.41	0.63	43.52	2.26	3.5
36.AI→BNB→USDT	262.06	8.46	170.41	10.35	98.73	6.04	3.0
37.DASH→ETH→USDT	90.40	0.21	88.95	1.12	50.19	1.36	3.5
38.HBAR→BNB→USDT	-100.00	-0.15	-100.00	-0.29	-30.82	-0.14	N/A
39.CHZ→BTC→USDT	-100.00	-0.35	-57.74	-1.26	-25.95	-1.11	N/A
40.ONT→USDC→USDT	159.36	4.29	139.12	7.41	82.06	9.21	3.5
41.CATI→BNB→USDT	-100.00	-1.38	-99.97	-2.46	-96.83	-9.79	N/A
42.BAT→BTC→USDT	76.04	2.34	51.50	3.72	29.89	3.81	3.5
43.GRT→EUR→USDT	269.51	9.37	161.51	10.46	87.30	6.94	3.0
44.BOME→EUR→USDT	768.24	21.18	704.23	23.75	558.72	13.41	3.0
45.NEIRO→EUR→USDT	9953.87	560.06	2017.48	157.52	305.10	21.96	2.5
46.RARE→USDC→USDT	260.45	6.16	169.29	7.25	78.87	9.27	3.5
47.ICP→EUR→USDT	209.77	6.76	142.45	7.25	68.25	5.14	3.0
48.IMX→BTC→USDT	281.98	22.23	173.97	18.86	75.81	15.30	2.5
49.RONIN→FDUSD→USDT	-100.00	-0.36	-100.00	-0.09	-54.40	-2.35	N/A
50.SUSHI→BTC→USDT	-58.14	-0.70	31.32	1.49	32.65	4.29	3.5
51.DIA→BTC→USDT	14.68	0.52	48.16	4.25	29.17	5.21	3.5
52.BNB→TUSD→USDT	-100.00	-0.28	-73.38	-2.26	-23.68	-2.34	N/A
53.YGG→USDC→USDT	184.31	4.57	142.83	7.52	70.53	10.02	3.5
54.SFP→BTC→USDT	-100.00	-0.34	-62.45	-1.72	-28.49	-2.12	N/A
55.ZRX→BTC→USDT	12.38	0.47	33.09	2.94	21.30	4.42	3.5
56.LTC→EUR→USDT	-100.00	-0.10	-100.00	-0.17	-62.39	-2.39	N/A
57.FLOKI→USDC→USDT	-100.00	-0.21	-100.00	-0.07	-100.00	-0.41	N/A
58.BTC→EUR→USDT	-100.00	-0.10	-100.00	-0.28	-47.51	-3.96	N/A
59.AAVE→ETH→USDT	-100.00	-0.22	-100.00	-0.33	-38.32	-5.00	N/A
60.MANA→ETH→USDT	-100.00	-0.21	-35.31	-2.21	3.00	0.34	3.5
61.DOGE→TUSD→USDT	101.66	4.88	88.51	9.48	48.62	11.93	3.5
62.AVAX→BNB→USDT	-76.53	-1.32	14.41	1.14	16.92	3.53	3.5
63.EOS→ETH→USDT	-38.81	-2.24	7.10	1.05	5.06	1.43	3.5
64.SXP→BTC→USDT	11.27	0.11	33.30	0.66	28.22	0.92	3.5

Complete parameter sensitivity results showing returns and Sharpe ratios across Z-entry threshold variations (2.5, 3.0, 3.5) for all 64 strategies. Baseline is Z=3.0. Higher Z-entry thresholds generally improve Sharpe ratios by focusing on stronger signals, though this may reduce absolute returns. Hub currency strategies (ETH, BTC, LTC, BNB) remain unprofitable across all parameter settings. The "Optimal Z" column indicates which threshold yields the highest Sharpe ratio for each strategy.

Table B7: Parameter Sensitivity Analysis - Z-Exit Threshold Variations

Strategy	0.15 Ret(%)	0.15 Sharpe	0.20 Ret(%)	0.20 Sharpe	0.25 Ret(%)	0.25 Sharpe	Optimal
01.XRP→TUSD→USDT	1.75	0.16	1.93	0.17	-5.13	-0.40	0.5
02.INJ→BNB→USDT	70.62	5.19	64.81	4.84	49.31	3.30	0.3
03.DOT→EUR→USDT	-2.65	-0.11	-0.95	-0.04	-0.39	-0.02	N/A
04.SUI→EUR→USDT	30.58	1.13	33.20	1.21	34.47	1.25	0.7
05.LINK→EUR→USDT	28.77	1.51	28.40	1.49	28.34	1.49	0.3
06.UNI→ETH→USDT	-87.96	-50.23	-89.49	-9.13	-91.82	-4.87	N/A
07.LISTA→BNB→USDT	235.53	11.89	215.10	10.64	190.25	9.32	0.3
08.LRC→ETH→USDT	32.70	1.93	25.77	1.54	12.54	0.62	0.3
09.IOTA→ETH→USDT	-47.17	-1.69	-56.52	-1.80	-67.62	-1.77	N/A
10.MANTA→USDC→USDT	110.32	6.69	113.76	6.73	116.00	6.82	0.7
11.VET→EUR→USDT	139.70	7.80	144.19	7.83	145.11	7.93	0.7
12.GALA→EUR→USDT	272.42	10.97	265.92	10.74	265.68	10.52	0.3
13.MOVE→BNB→USDT	92.08	3.00	84.45	2.63	79.62	2.52	0.3
14.POL→EUR→USDT	130.78	13.71	138.00	14.74	136.80	13.80	0.5
15.PNUT→EUR→USDT	980.46	13.30	965.28	12.88	1005.22	13.52	0.7
16.AGLD→BTC→USDT	132.92	6.01	132.55	5.95	126.29	5.72	0.3
17.REZ→USDC→USDT	253.56	9.02	259.99	9.25	261.77	9.20	0.5
18.XLM→EUR→USDT	47.96	2.15	51.02	2.28	51.83	2.33	0.7
19.UMA→BTC→USDT	48.13	1.68	69.57	2.45	62.66	2.18	0.5
20.TRX→EUR→USDT	-100.00	-0.28	-100.00	-0.32	-100.00	-0.28	N/A
21.ARKM→BNB→USDT	264.82	12.88	295.47	12.47	284.48	11.60	0.3
22.STEEM→ETH→USDT	43.90	3.86	42.83	3.34	38.15	2.94	0.3
23.CELR→BTC→USDT	4461.18	48.49	4812.64	51.89	5388.59	66.06	0.7
24.WIF→EUR→USDT	326.31	10.55	324.72	10.12	322.38	10.18	0.3
25.ADX→ETH→USDT	1.88	0.09	1.89	0.09	-1.42	-0.07	0.3
26.1000SATS→USDC→USDT	58.10	2.36	59.92	2.37	58.46	2.29	0.5
27.FET→BNB→USDT	128.21	5.41	127.73	5.44	143.86	4.70	0.5
28.ETH→EUR→USDT	-44.10	-1.65	-47.95	-1.84	-53.41	-2.04	N/A
29.SOL→TUSD→USDT	39.92	3.08	37.62	2.85	36.34	2.69	0.3
30.MEME→USDC→USDT	149.39	4.45	156.81	4.57	157.65	4.62	0.7
31.TFUEL→BTC→USDT	340.24	15.00	326.29	15.11	291.60	11.74	0.5
32.WIN→EUR→USDT	-100.00	-0.32	-100.00	-0.06	-100.00	-0.28	N/A
33.CTK→BNB→USDT	96.53	7.38	98.62	7.09	91.76	6.78	0.3
34.ENA→BNB→USDT	281.99	10.82	280.30	10.40	283.26	9.86	0.3
35.PYTH→USDC→USDT	24.99	0.67	24.41	0.63	25.73	0.67	0.3
36.AI→BNB→USDT	177.26	9.84	170.41	10.35	152.01	8.55	0.5
37.DASH→ETH→USDT	97.24	1.38	88.95	1.12	73.57	0.82	0.3
38.HBAR→BNB→USDT	-100.00	-0.20	-100.00	-0.29	-100.00	-0.34	N/A
39.CHZ→BNB→USDT	-44.98	-1.17	-57.74	-1.26	-71.22	-1.26	N/A
40.ONT→USDC→USDT	139.66	7.25	139.12	7.41	137.77	7.79	0.7
41.CATI→BNB→USDT	-99.96	-2.36	-99.97	-2.46	-99.92	-2.55	N/A
42.BAT→BTC→USDT	59.15	4.44	51.50	3.72	44.43	2.91	0.3
43.GRT→EUR→USDT	167.94	10.34	161.51	10.46	156.00	9.55	0.5
44.BOME→EUR→USDT	626.87	23.48	704.23	23.75	688.74	21.07	0.5
45.NEIRO→EUR→USDT	2377.57	190.37	2017.48	157.52	1885.10	186.21	0.3
46.RARE→USDC→USDT	163.25	6.96	169.29	7.25	169.79	7.26	0.7
47.ICP→EUR→USDT	145.13	7.50	142.45	7.25	134.33	6.20	0.3
48.IMX→BTC→USDT	170.54	18.83	173.97	18.86	167.28	17.61	0.5
49.RONIN→FDUSD→USDT	-100.00	-0.30	-100.00	-0.09	-100.00	-0.10	N/A
50.SUSHI→BTC→USDT	30.47	1.46	31.32	1.49	29.83	1.40	0.5
51.DIA→BTC→USDT	50.27	4.66	48.16	4.25	42.36	3.34	0.3
52.BNB→TUSD→USDT	-71.02	-2.37	-73.38	-2.26	-76.61	-2.05	N/A
53.YGG→USDC→USDT	139.82	7.22	142.83	7.52	142.70	7.55	0.7
54.SFP→BTC→USDT	-53.42	-1.61	-62.45	-1.72	-76.22	-1.62	N/A
55.ZRX→BTC→USDT	37.05	3.35	33.09	2.94	23.02	2.02	0.3
56.LTC→EUR→USDT	-100.00	-0.23	-100.00	-0.17	-100.00	-0.09	N/A
57.FLOKI→USDC→USDT	-100.00	-0.30	-100.00	-0.07	-100.00	-0.22	N/A
58.BTC→EUR→USDT	-100.00	-0.14	-100.00	-0.28	-100.00	-0.27	N/A
59.AAVE→ETH→USDT	-100.00	-0.39	-100.00	-0.33	-100.00	-0.31	N/A
60.MANA→ETH→USDT	-28.63	-1.96	-35.31	-2.21	-42.20	-2.28	N/A
61.DOGE→TUSD→USDT	91.67	11.31	88.51	9.48	72.63	8.38	0.3
62.AVAX→BNB→USDT	14.07	1.11	14.41	1.14	26.00	1.20	0.7
63.EOS→ETH→USDT	12.80	2.02	7.10	1.05	3.31	0.44	0.3
64.SXP→BTC→USDT	40.35	0.89	33.30	0.66	20.16	0.35	0.3

Returns and Sharpe ratios across z-exit threshold variations for all 64 strategies. Baseline values are indicated in the table. "Optimal" is the value with highest Sharpe; N/A if all are negative.

Table B8: Parameter Sensitivity Analysis - Lookback Window Variations

Strategy	45min Ret(%)	45min Sharpe	60min Ret(%)	60min Sharpe	90min Ret(%)	90min Sharpe	Optimal (min)
01.XRP→TUSD→USDT	-24.01	-1.36	1.93	0.17	33.85	4.48	90
02.INJ→BNB→USDT	13.67	0.70	64.81	4.84	125.69	9.57	90
03.DOT→EUR→USDT	-14.56	-0.55	-0.95	-0.04	15.45	0.69	90
04.SUI→EUR→USDT	12.22	0.46	33.20	1.21	49.89	1.75	90
05.LINK→EUR→USDT	17.72	0.99	28.40	1.49	40.30	2.06	90
06.UNI→ETH→USDT	-96.98	-1.41	-89.49	-9.13	-73.41	-6.76	N/A
07.LISTA→BNB→USDT	125.32	4.59	215.10	10.64	333.90	16.20	90
08.LRC→ETH→USDT	-28.44	-0.95	25.77	1.54	78.46	7.41	90
09.IOTA→ETH→USDT	-100.00	-0.15	-56.52	-1.80	20.25	0.98	90
10.MANTA→USDC→USDT	85.67	5.81	113.76	6.73	131.60	7.91	90
11.VET→EUR→USDT	105.39	6.40	144.19	7.83	179.77	9.60	90
12.GALA→EUR→USDT	199.92	11.04	265.92	10.74	348.25	12.48	90
13.MOVE→BNB→USDT	31.94	0.90	84.45	2.63	154.90	5.25	90
14.POL→EUR→USDT	103.86	18.02	138.00	14.74	182.41	14.45	45
15.PNUT→EUR→USDT	840.38	12.75	965.28	12.88	992.84	11.63	60
16.AGLD→BTC→USDT	90.81	4.75	132.55	5.95	170.15	7.28	90
17.REZ→USDC→USDT	211.71	8.86	259.99	9.25	311.13	9.90	90
18.XLM→EUR→USDT	24.18	1.17	51.02	2.28	72.73	3.09	90
19.UMA→BTC→USDT	26.26	0.64	69.57	2.45	95.61	2.56	90
20.TRX→EUR→USDT	-100.00	-0.34	-100.00	-0.32	-100.00	-0.29	N/A
21.ARKM→BNB→USDT	200.27	9.84	295.47	12.47	341.91	13.36	90
22.STEEM→ETH→USDT	11.01	0.73	42.83	3.34	69.92	6.41	90
23.CELR→BTC→USDT	5272.71	61.83	4812.64	51.89	5013.09	70.77	90
24.WIF→EUR→USDT	287.83	10.47	324.72	10.12	393.26	10.63	90
25.ADX→ETH→USDT	-6.67	-0.38	1.89	0.09	12.99	0.55	90
26.1000SATS→USDC→USDT	44.45	1.99	59.92	2.37	78.27	2.63	90
27.FET→BNB→USDT	73.85	4.49	127.73	5.44	176.39	6.34	90
28.ETH→EUR→USDT	-63.31	-2.69	-47.95	-1.84	-38.03	-1.26	N/A
29.SOL→TUSD→USDT	6.68	0.46	37.62	2.85	71.40	6.00	90
30.MEME→USDC→USDT	117.41	3.75	156.81	4.57	190.55	5.25	90
31.TFUEL→BTC→USDT	276.38	11.73	326.29	15.11	417.17	16.89	90
32.WIN→EUR→USDT	-100.00	-0.17	-100.00	-0.06	-100.00	-0.00	N/A
33.CTK→BNB→USDT	54.28	4.04	98.62	7.09	163.83	11.21	90
34.ENA→BNB→USDT	189.04	8.41	280.30	10.40	328.10	12.53	90
35.PYTH→USDC→USDT	16.28	0.46	24.41	0.63	44.32	1.04	90
36.AI→BNB→USDT	110.75	5.54	170.41	10.35	220.59	10.54	90
37.DASH→ETH→USDT	27.18	0.24	88.95	1.12	150.30	2.86	90
38.HBAR→BNB→USDT	-100.00	-0.30	-100.00	-0.29	-100.00	-0.23	N/A
39.CHZ→BNB→USDT	-100.00	-0.01	-57.74	-1.26	8.97	0.33	90
40.ONT→USDC→USDT	93.45	6.46	139.12	7.41	184.08	8.09	90
41.CATI→BNB→USDT	-99.99	-2.50	-99.97	-2.46	-99.54	-2.28	N/A
42.BAT→BTC→USDT	10.58	0.56	51.50	3.72	96.80	9.88	90
43.GRT→EUR→USDT	112.08	6.47	161.51	10.46	225.59	12.12	90
44.BOME→EUR→USDT	580.68	12.95	704.23	23.75	705.45	19.56	60
45.NEIRO→EUR→USDT	1382.53	92.19	2017.48	157.52	3929.18	351.80	90
46.RARE→USDC→USDT	139.19	7.06	169.29	7.25	195.32	7.63	90
47.ICP→EUR→USDT	89.70	3.68	142.45	7.25	205.42	11.38	90
48.IMX→BTC→USDT	125.83	18.11	173.97	18.86	207.12	18.49	60
49.RONIN→FDUSD→USDT	-100.00	-0.51	-100.00	-0.09	-76.29	-1.22	N/A
50.SUSHI→BTC→USDT	15.41	0.84	31.32	1.49	51.03	2.24	90
51.DIA→BTC→USDT	23.15	1.74	48.16	4.25	74.74	7.12	90
52.BNB→TUSD→USDT	-100.00	-0.04	-73.38	-2.26	-40.50	-2.10	N/A
53.YGG→USDC→USDT	110.80	6.82	142.83	7.52	174.36	7.57	90
54.SFP→BTC→USDT	-100.00	-0.06	-62.45	-1.72	1.30	0.09	90
55.ZRX→BTC→USDT	-10.23	-0.87	33.09	2.94	70.19	5.39	90
56.LTC→EUR→USDT	-100.00	-0.24	-100.00	-0.17	-100.00	-0.12	N/A
57.FLOKI→USDC→USDT	-100.00	-0.16	-100.00	-0.07	-100.00	-0.02	N/A
58.BTC→EUR→USDT	-100.00	-0.28	-100.00	-0.28	-100.00	-0.15	N/A
59.AAVE→ETH→USDT	-100.00	-0.06	-100.00	-0.33	-100.00	-0.37	N/A
60.MANA→ETH→USDT	-70.97	-2.01	-35.31	-2.21	21.11	1.23	90
61.DOGE→TUSD→USDT	22.92	2.42	88.51	9.48	136.06	13.37	90
62.AVAX→BNB→USDT	-10.05	-0.81	14.41	1.14	40.67	3.08	90
63.EOS→ETH→USDT	-21.83	-2.06	7.10	1.05	45.89	8.46	90
64.SXP→BTC→USDT	-16.99	-0.20	33.30	0.66	91.89	2.94	90

Returns and Sharpe ratios across lookback window variations (45, 60, 90 minutes) for all 64 strategies. Baseline is 60 minutes. "Optimal" is the window with highest Sharpe; N/A if all are negative.

Table B9: Parameter Sensitivity Analysis - Position Size Variations

Strategy	20% Ret(%)	20% Sharpe	30% Ret(%)	30% Sharpe	40% Ret(%)	40% Sharpe	Optimal
01.XRP→TUSD→USDT	1.36	0.18	1.93	0.17	2.45	0.17	20%
02.INJ→BNB→USDT	43.14	4.55	64.81	4.84	86.60	5.10	40%
03.DOT→EUR→USDT	-0.55	-0.04	-0.95	-0.04	-1.40	-0.04	N/A
04.SUI→EUR→USDT	22.11	1.13	33.20	1.21	44.32	1.29	40%
05.LINK→EUR→USDT	19.36	1.45	28.40	1.49	37.16	1.52	40%
06.UNI→ETH→USDT	-60.76	-11.77	-89.49	-9.13	-100.00	-0.52	N/A
07.LISTA→BNB→USDT	141.00	9.09	215.10	10.64	290.90	11.91	40%
08.LRC→ETH→USDT	17.16	1.48	25.77	1.54	34.40	1.59	40%
09.IOTA→ETH→USDT	-38.11	-2.06	-56.52	-1.80	-74.35	-1.47	N/A
10.MANTA→USDC→USDT	75.25	5.79	113.76	6.73	152.74	7.62	40%
11.VET→EUR→USDT	95.40	6.97	144.19	7.83	193.50	8.55	40%
12.GALA→EUR→USDT	175.56	10.10	265.92	10.74	357.32	11.28	40%
13.MOVE→BNB→USDT	56.08	2.34	84.45	2.63	112.98	2.90	40%
14.POL→EUR→USDT	91.25	15.42	138.00	14.74	185.31	14.34	20%
15.PNUT→EUR→USDT	635.56	10.94	965.28	12.88	1297.65	14.65	40%
16.AGLD→BTC→USDT	88.22	5.75	132.55	5.95	176.97	6.14	40%
17.REZ→USDC→USDT	171.78	8.72	259.99	9.25	349.08	9.78	40%
18.XLM→EUR→USDT	33.89	2.12	51.02	2.28	68.26	2.43	40%
19.UMA→BTC→USDT	45.89	2.15	69.57	2.45	93.76	2.80	40%
20.TRX→EUR→USDT	-100.00	-0.35	-100.00	-0.32	-100.00	-0.10	N/A
21.ARKM→BNB→USDT	194.71	11.19	295.47	12.47	397.70	13.43	40%
22.STEEM→ETH→USDT	28.55	3.20	42.83	3.34	57.10	3.47	40%
23.CELR→BTC→USDT	1947.12	30.16	4812.64	51.89	9683.35	81.73	40%
24.WIF→EUR→USDT	214.24	9.72	324.72	10.12	436.47	10.58	40%
25.ADX→ETH→USDT	1.26	0.09	1.89	0.09	2.52	0.09	20%
26.1000SATS→USDC→USDT	39.84	2.25	59.92	2.37	80.13	2.48	40%
27.FET→BNB→USDT	84.51	4.65	127.73	5.44	171.46	6.20	40%
28.ETH→EURI→USDT	-32.14	-1.97	-47.95	-1.84	-63.50	-1.65	N/A
29.SOL→TUSD→USDT	25.10	2.73	37.62	2.85	50.18	2.97	40%
30.MEME→USDC→USDT	103.72	4.14	156.81	4.57	210.47	4.96	40%
31.TFUEL→BTC→USDT	216.14	13.33	326.29	15.11	436.89	16.30	40%
32.WIN→EUR→USDT	-100.00	-0.26	-100.00	-0.06	-100.00	-0.06	N/A
33.CTK→BNB→USDT	65.31	6.44	98.62	7.09	132.28	7.64	40%
34.ENA→BNB→USDT	184.86	8.91	280.30	10.40	377.01	11.62	40%
35.PYTH→USDC→USDT	16.26	0.58	24.41	0.63	32.57	0.68	40%
36.AI→BNB→USDT	112.81	9.77	170.41	10.35	228.52	10.72	40%
37.DASH→ETH→USDT	59.04	1.11	88.95	1.12	119.11	1.12	30%
38.HBAR→BNB→USDT	-100.00	-0.22	-100.00	-0.29	-100.00	-0.23	N/A
39.CHZ→BNB→USDT	-38.82	-1.56	-57.74	-1.26	-76.15	-0.86	N/A
40.ONT→USDC→USDT	92.22	7.11	139.12	7.41	186.35	7.67	40%
41.CATI→BNB→USDT	-99.39	-3.87	-99.97	-2.46	-100.00	-1.73	N/A
42.BAT→BTC→USDT	34.42	3.46	51.50	3.72	68.56	3.96	40%
43.GRT→EUR→USDT	106.75	9.89	161.51	10.46	216.93	10.87	40%
44.BOME→EUR→USDT	438.97	16.88	704.23	23.75	991.23	29.79	40%
45.NEIRO→EUR→USDT	775.41	72.08	2017.48	157.52	4555.50	339.14	40%
46.RARE→USDC→USDT	111.86	6.78	169.29	7.25	227.44	7.67	40%
47.ICP→EUR→USDT	94.34	6.11	142.45	7.25	191.03	8.28	40%
48.IMX→BTC→USDT	115.24	18.74	173.97	18.86	233.25	18.76	30%
49.RONIN→FDUSD→USDT	-74.83	-1.91	-100.00	-0.09	-100.00	-0.02	N/A
50.SUSHI→BTC→USDT	21.14	1.51	31.32	1.49	41.38	1.47	20%
51.DIA→BTC→USDT	32.21	3.94	48.16	4.25	64.09	4.56	40%
52.BNB→TUSD→USDT	-49.48	-3.44	-73.38	-2.26	-95.91	-0.68	N/A
53.YGG→USDC→USDT	94.45	7.24	142.83	7.52	191.79	7.80	40%
54.SFP→BTC→USDT	-41.85	-2.28	-62.45	-1.72	-82.46	-1.10	N/A
55.ZRX→BTC→USDT	22.13	2.90	33.09	2.94	44.04	2.98	40%
56.LTC→EUR→USDT	-100.00	-0.09	-100.00	-0.17	-100.00	-0.02	N/A
57.FLOKI→USDC→USDT	-100.00	-0.23	-100.00	-0.07	-100.00	-0.02	N/A
58.BTC→EURI→USDT	-97.18	-0.55	-100.00	-0.28	-100.00	-0.30	N/A
59.AAVE→ETH→USDT	-90.48	-1.15	-100.00	-0.33	-100.00	-0.33	N/A
60.MANA→ETH→USDT	-23.61	-2.39	-35.31	-2.21	-46.92	-2.00	N/A
61.DOGE→TUSD→USDT	58.14	9.19	88.51	9.48	119.61	9.79	40%
62.AVAX→BNB→USDT	9.83	1.13	14.41	1.14	18.93	1.15	40%
63.EOS→ETH→USDT	4.72	1.05	7.10	1.05	9.49	1.05	40%
64.SXP→BTC→USDT	22.34	0.65	33.30	0.66	44.21	0.66	40%

Returns and Sharpe ratios across position size variations (20%, 30%, 40%) for all 64 strategies. Baseline is 30%.

“Optimal” is the size with highest Sharpe; N/A if all are negative.

Table B10: Parameter Sensitivity Analysis - Centrality Weight Variations

Strategy	0.5 Ret(%)	0.5 Sharpe	0.6 Ret(%)	0.6 Sharpe	0.7 Ret(%)	0.7 Sharpe	0.8 Ret(%)	0.8 Sharpe	Optimal
01.XRP→TUSD→USDT	1.93	0.17	1.93	0.17	1.93	0.17	1.93	0.17	0.5
02.INJ→BNB→USDT	64.81	4.84	64.81	4.84	64.81	4.84	64.81	4.84	0.5
03.DOT→EUR→USDT	-0.95	-0.04	-0.95	-0.04	-0.95	-0.04	-0.95	-0.04	N/A
04.SUI→EUR→USDT	33.20	1.21	33.20	1.21	33.20	1.21	33.20	1.21	0.5
05.LINK→EUR→USDT	28.40	1.49	28.40	1.49	28.40	1.49	28.40	1.49	0.5
06.UNI→ETH→USDT	-89.49	-9.13	-89.49	-9.13	-89.49	-9.13	-89.49	-9.13	N/A
07.LISTA→BNB→USDT	215.10	10.64	215.10	10.64	215.10	10.64	215.10	10.64	0.5
08.LRC→ETH→USDT	25.77	1.54	25.77	1.54	25.77	1.54	25.77	1.54	0.5
09.IOTA→ETH→USDT	-56.52	-1.80	-56.52	-1.80	-56.52	-1.80	-56.52	-1.80	N/A
10.MANTA→USDC→USDT	113.76	6.73	113.76	6.73	113.76	6.73	113.76	6.73	0.5
11.VET→EUR→USDT	144.19	7.83	144.19	7.83	144.19	7.83	144.19	7.83	0.5
12.GALA→EUR→USDT	265.92	10.74	265.92	10.74	265.92	10.74	265.92	10.74	0.5
13.MOVE→BNB→USDT	84.45	2.63	84.45	2.63	84.45	2.63	84.45	2.63	0.5
14.POL→EUR→USDT	138.00	14.74	138.00	14.74	138.00	14.74	138.00	14.74	0.5
15.PNUT→EUR→USDT	965.28	12.88	965.28	12.88	965.28	12.88	965.28	12.88	0.5
16.AGLD→BTC→USDT	132.55	5.95	132.55	5.95	132.55	5.95	132.55	5.95	0.5
17.REZ→USDC→USDT	259.99	9.25	259.99	9.25	259.99	9.25	259.99	9.25	0.5
18.XLM→EUR→USDT	51.02	2.28	51.02	2.28	51.02	2.28	51.02	2.28	0.5
19.UMA→BTC→USDT	69.57	2.45	69.57	2.45	69.57	2.45	69.57	2.45	0.5
20.TRX→EUR→USDT	-100.00	-0.32	-100.00	-0.32	-100.00	-0.32	-100.00	-0.32	N/A
21.ARKM→BNB→USDT	295.47	12.47	295.47	12.47	295.47	12.47	295.47	12.47	0.5
22.STEEM→ETH→USDT	42.83	3.34	42.83	3.34	42.83	3.34	42.83	3.34	0.5
23.CELR→BTC→USDT	4812.64	51.89	4812.64	51.89	4812.64	51.89	4812.64	51.89	0.5
24.WIF→EUR→USDT	324.72	10.12	324.72	10.12	324.72	10.12	324.72	10.12	0.5
25.ADX→ETH→USDT	1.89	0.09	1.89	0.09	1.89	0.09	1.89	0.09	0.5
26.1000SATS→USDC→USDT	59.92	2.37	59.92	2.37	59.92	2.37	59.92	2.37	0.5
27.FET→BNB→USDT	127.73	5.44	127.73	5.44	127.73	5.44	127.73	5.44	0.5
28.ETH→EUR→USDT	-47.95	-1.84	-47.95	-1.84	-47.95	-1.84	-47.95	-1.84	N/A
29.SOL→TUSD→USDT	37.62	2.85	37.62	2.85	37.62	2.85	37.62	2.85	0.5
30.MEME→USDC→USDT	156.81	4.57	156.81	4.57	156.81	4.57	156.81	4.57	0.5
31.TFUEL→BTC→USDT	326.29	15.11	326.29	15.11	326.29	15.11	326.29	15.11	0.5
32.WIN→EUR→USDT	-100.00	-0.06	-100.00	-0.06	-100.00	-0.06	-100.00	-0.06	N/A
33.CTK→BNB→USDT	98.62	7.09	98.62	7.09	98.62	7.09	98.62	7.09	0.5
34.ENA→BNB→USDT	280.30	10.40	280.30	10.40	280.30	10.40	280.30	10.40	0.5
35.PYTH→USDC→USDT	24.41	0.63	24.41	0.63	24.41	0.63	24.41	0.63	0.5
36.AI→BNB→USDT	170.41	10.35	170.41	10.35	170.41	10.35	170.41	10.35	0.5
37.DASH→ETH→USDT	88.95	1.12	88.95	1.12	88.95	1.12	88.95	1.12	0.5
38.HBAR→BNB→USDT	-100.00	-0.29	-100.00	-0.29	-100.00	-0.29	-100.00	-0.29	N/A
39.CHZ→BNB→USDT	-57.74	-1.26	-57.74	-1.26	-57.74	-1.26	-57.74	-1.26	N/A
40.ONT→USDC→USDT	139.12	7.41	139.12	7.41	139.12	7.41	139.12	7.41	0.5
41.CATI→BNB→USDT	-99.97	-2.46	-99.97	-2.46	-99.97	-2.46	-99.97	-2.46	N/A
42.BAT→BTC→USDT	51.50	3.72	51.50	3.72	51.50	3.72	51.50	3.72	0.5
43.GRT→EUR→USDT	161.51	10.46	161.51	10.46	161.51	10.46	161.51	10.46	0.5
44.BOME→EUR→USDT	704.23	23.75	704.23	23.75	704.23	23.75	704.23	23.75	0.5
45.NEIRO→EUR→USDT	2017.48	157.52	2017.48	157.52	2017.48	157.52	2017.48	157.52	0.5
46.RARE→USDC→USDT	169.29	7.25	169.29	7.25	169.29	7.25	169.29	7.25	0.5
47.ICP→EUR→USDT	142.45	7.25	142.45	7.25	142.45	7.25	142.45	7.25	0.5
48.IMX→BTC→USDT	173.97	18.86	173.97	18.86	173.97	18.86	173.97	18.86	0.5
49.ROBIN→FDUSD→USDT	-100.00	-0.09	-100.00	-0.09	-100.00	-0.09	-100.00	-0.09	N/A
50.SUSHI→BTC→USDT	31.32	1.49	31.32	1.49	31.32	1.49	31.32	1.49	0.5
51.DIA→BTC→USDT	48.16	4.25	48.16	4.25	48.16	4.25	48.16	4.25	0.5
52.BNB→TUSD→USDT	-73.38	-2.26	-73.38	-2.26	-73.38	-2.26	-73.38	-2.26	N/A
53.YGG→USDC→USDT	142.83	7.52	142.83	7.52	142.83	7.52	142.83	7.52	0.5
54.SFP→BTC→USDT	-62.45	-1.72	-62.45	-1.72	-62.45	-1.72	-62.45	-1.72	N/A
55.ZRX→BTC→USDT	33.09	2.94	33.09	2.94	33.09	2.94	33.09	2.94	0.5
56.LTC→EUR→USDT	-100.00	-0.17	-100.00	-0.17	-100.00	-0.17	-100.00	-0.17	N/A
57.FLOKI→USDC→USDT	-100.00	-0.07	-100.00	-0.07	-100.00	-0.07	-100.00	-0.07	N/A
58.BTC→EUR→USDT	-100.00	-0.28	-100.00	-0.28	-100.00	-0.28	-100.00	-0.28	N/A
59.AAVE→ETH→USDT	-100.00	-0.33	-100.00	-0.33	-100.00	-0.33	-100.00	-0.33	N/A
60.MANA→ETH→USDT	-35.31	-2.21	-35.31	-2.21	-35.31	-2.21	-35.31	-2.21	N/A
61.DOGE→TUSD→USDT	88.51	9.48	88.51	9.48	88.51	9.48	88.51	9.48	0.5
62.AVAX→BNB→USDT	14.41	1.14	14.41	1.14	14.41	1.14	14.41	1.14	0.5
63.EOS→ETH→USDT	7.10	1.05	7.10	1.05	7.10	1.05	7.10	1.05	0.5
64.SXP→BTC→USDT	33.30	0.66	33.30	0.66	33.30	0.66	33.30	0.66	0.5

Returns and Sharpe ratios across centrality weight variations for all 64 strategies. Baseline values are indicated in the table. "Optimal" is the value with highest Sharpe; N/A if all are negative.

Table B11: Parameter Sensitivity Analysis - κ BASE Variations

Strategy	5e-5 Ret(%)	5e-5 Sharpe	1e-4 Ret(%)	1e-4 Sharpe	2e-4 Ret(%)	2e-4 Sharpe	Optimal
01.XRP→TUSD→USDT	1.93	0.17	1.93	0.17	1.93	0.17	5e-05
02.INJ→BNB→USDT	64.81	4.84	64.81	4.84	64.81	4.84	5e-05
03.DOT→EUR→USDT	-0.95	-0.04	-0.95	-0.04	-0.95	-0.04	N/A
04.SUI→EUR→USDT	33.20	1.21	33.20	1.21	33.20	1.21	5e-05
05.LINK→EUR→USDT	28.40	1.49	28.40	1.49	28.40	1.49	5e-05
06.UNI→ETH→USDT	-89.49	-9.13	-89.49	-9.13	-89.49	-9.13	N/A
07.LISTA→BNB→USDT	215.10	10.64	215.10	10.64	215.10	10.64	5e-05
08.LRC→ETH→USDT	25.77	1.54	25.77	1.54	25.77	1.54	5e-05
09.IOTA→ETH→USDT	-56.52	-1.80	-56.52	-1.80	-56.52	-1.80	N/A
10.MANTA→USDC→USDT	113.76	6.73	113.76	6.73	113.76	6.73	5e-05
11.VET→EUR→USDT	144.19	7.83	144.19	7.83	144.19	7.83	5e-05
12.GALA→EUR→USDT	265.92	10.74	265.92	10.74	265.92	10.74	5e-05
13.MOVE→BNB→USDT	84.45	2.63	84.45	2.63	84.45	2.63	5e-05
14.POL→EUR→USDT	138.00	14.74	138.00	14.74	138.00	14.74	5e-05
15.PNUT→EUR→USDT	965.28	12.88	965.28	12.88	965.28	12.88	5e-05
16.AGLD→BTC→USDT	132.55	5.95	132.55	5.95	132.55	5.95	5e-05
17.REZ→USDC→USDT	259.99	9.25	259.99	9.25	259.99	9.25	5e-05
18.XLM→EUR→USDT	51.02	2.28	51.02	2.28	51.02	2.28	5e-05
19.UMA→BTC→USDT	69.57	2.45	69.57	2.45	69.57	2.45	5e-05
20.TRX→EUR→USDT	-100.00	-0.32	-100.00	-0.32	-100.00	-0.32	N/A
21.ARKM→BNB→USDT	295.47	12.47	295.47	12.47	295.47	12.47	5e-05
22.STEEM→ETH→USDT	42.83	3.34	42.83	3.34	42.83	3.34	5e-05
23.CELR→BTC→USDT	4812.64	51.89	4812.64	51.89	4812.64	51.89	5e-05
24.WIF→EUR→USDT	324.72	10.12	324.72	10.12	324.72	10.12	5e-05
25.ADX→ETH→USDT	1.89	0.09	1.89	0.09	1.89	0.09	5e-05
26.1000SATS→USDC→USDT	59.92	2.37	59.92	2.37	59.92	2.37	5e-05
27.FET→BNB→USDT	127.73	5.44	127.73	5.44	127.73	5.44	5e-05
28.ETH→EURI→USDT	-47.95	-1.84	-47.95	-1.84	-47.95	-1.84	N/A
29.SOL→TUSD→USDT	37.62	2.85	37.62	2.85	37.62	2.85	5e-05
30.MEME→USDC→USDT	156.81	4.57	156.81	4.57	156.81	4.57	5e-05
31.TFUEL→BTC→USDT	326.29	15.11	326.29	15.11	326.29	15.11	5e-05
32.WIN→EUR→USDT	-100.00	-0.06	-100.00	-0.06	-100.00	-0.06	N/A
33.CTK→BNB→USDT	98.62	7.09	98.62	7.09	98.62	7.09	5e-05
34.ENA→BNB→USDT	280.30	10.40	280.30	10.40	280.30	10.40	5e-05
35.PYTH→USDC→USDT	24.41	0.63	24.41	0.63	24.41	0.63	5e-05
36.AI→BNB→USDT	170.41	10.35	170.41	10.35	170.41	10.35	5e-05
37.DASH→ETH→USDT	88.95	1.12	88.95	1.12	88.95	1.12	5e-05
38.HBAR→BNB→USDT	-100.00	-0.29	-100.00	-0.29	-100.00	-0.29	N/A
39.CHZ→BNB→USDT	-57.74	-1.26	-57.74	-1.26	-57.74	-1.26	N/A
40.ONT→USDC→USDT	139.12	7.41	139.12	7.41	139.12	7.41	5e-05
41.CATI→BNB→USDT	-99.97	-2.46	-99.97	-2.46	-99.97	-2.46	N/A
42.BAT→BTC→USDT	51.50	3.72	51.50	3.72	51.50	3.72	5e-05
43.GRT→EUR→USDT	161.51	10.46	161.51	10.46	161.51	10.46	5e-05
45.NEIRO→EUR→USDT	2017.48	157.52	2017.48	157.52	2017.48	157.52	5e-05
46.RARE→USDC→USDT	169.29	7.25	169.29	7.25	169.29	7.25	5e-05
47.ICP→EUR→USDT	142.45	7.25	142.45	7.25	142.45	7.25	5e-05
48.IMX→BTC→USDT	173.97	18.86	173.97	18.86	173.97	18.86	5e-05
49.RONIN→FDUSD→USDT	-100.00	-0.09	-100.00	-0.09	-100.00	-0.09	N/A
50.SUSHI→BTC→USDT	31.32	1.49	31.32	1.49	31.32	1.49	5e-05
51.DIA→BTC→USDT	48.16	4.25	48.16	4.25	48.16	4.25	5e-05
52.BNB→TUSD→USDT	-73.38	-2.26	-73.38	-2.26	-73.38	-2.26	N/A
53.YGG→USDC→USDT	142.83	7.52	142.83	7.52	142.83	7.52	5e-05
54.SFP→BTC→USDT	-62.45	-1.72	-62.45	-1.72	-62.45	-1.72	N/A
55.ZRX→BTC→USDT	33.09	2.94	33.09	2.94	33.09	2.94	5e-05
56.LTC→EUR→USDT	-100.00	-0.17	-100.00	-0.17	-100.00	-0.17	N/A
57.FLOKI→USDC→USDT	-100.00	-0.07	-100.00	-0.07	-100.00	-0.07	N/A
58.BTC→EURI→USDT	-100.00	-0.28	-100.00	-0.28	-100.00	-0.28	N/A
59.AAVE→ETH→USDT	-100.00	-0.33	-100.00	-0.33	-100.00	-0.33	N/A
60.MANA→ETH→USDT	-35.31	-2.21	-35.31	-2.21	-35.31	-2.21	N/A
62.AVAX→BNB→USDT	14.41	1.14	14.41	1.14	14.41	1.14	5e-05
63.EOS→ETH→USDT	7.10	1.05	7.10	1.05	7.10	1.05	5e-05
64.SXP→BTC→USDT	33.30	0.66	33.30	0.66	33.30	0.66	5e-05

Returns and Sharpe ratios across κ BASE variations for all 64 strategies. Baseline values are indicated in the table.

“Optimal” is the value with highest Sharpe; N/A if all are negative.

Table B12: Parameter Sensitivity Analysis - κ α_{edge} Variations

Strategy	5e-4 Ret(%)	5e-4 Sharpe	1e-3 Ret(%)	1e-3 Sharpe	2e-3 Ret(%)	2e-3 Sharpe	Optimal
01.XRP→TUSD→USDT	1.93	0.17	1.93	0.17	1.93	0.17	0.0005
02.INJ→BNB→USDT	64.81	4.84	64.81	4.84	64.81	4.84	0.0005
03.DOT→EUR→USDT	-0.95	-0.04	-0.95	-0.04	-0.95	-0.04	N/A
04.SUI→EUR→USDT	33.20	1.21	33.20	1.21	33.20	1.21	0.0005
05.LINK→EUR→USDT	28.40	1.49	28.40	1.49	28.40	1.49	0.0005
06.UNI→ETH→USDT	-89.49	-9.13	-89.49	-9.13	-89.49	-9.13	N/A
07.LISTA→BNB→USDT	215.10	10.64	215.10	10.64	215.10	10.64	0.0005
08.LRC→ETH→USDT	25.77	1.54	25.77	1.54	25.77	1.54	0.0005
09.IOTA→ETH→USDT	-56.52	-1.80	-56.52	-1.80	-56.52	-1.80	N/A
10.MANTA→USDC→USDT	113.76	6.73	113.76	6.73	113.76	6.73	0.0005
11.VET→EUR→USDT	144.19	7.83	144.19	7.83	144.19	7.83	0.0005
12.GALA→EUR→USDT	265.92	10.74	265.92	10.74	265.92	10.74	0.0005
13.MOVE→BNB→USDT	84.45	2.63	84.45	2.63	84.45	2.63	0.0005
14.POL→EUR→USDT	138.00	14.74	138.00	14.74	138.00	14.74	0.0005
15.PNUT→EUR→USDT	965.28	12.88	965.28	12.88	965.28	12.88	0.0005
16.AGLD→BTC→USDT	132.55	5.95	132.55	5.95	132.55	5.95	0.0005
17.REZ→USDC→USDT	259.99	9.25	259.99	9.25	259.99	9.25	0.0005
18.XLM→EUR→USDT	51.02	2.28	51.02	2.28	51.02	2.28	0.0005
19.UMA→BTC→USDT	69.57	2.45	69.57	2.45	69.57	2.45	0.0005
20.TRX→EUR→USDT	-100.00	-0.32	-100.00	-0.32	-100.00	-0.32	N/A
21.ARKM→BNB→USDT	295.47	12.47	295.47	12.47	295.47	12.47	0.0005
22.STEEM→ETH→USDT	42.83	3.34	42.83	3.34	42.83	3.34	0.0005
23.CELR→BTC→USDT	4812.64	51.89	4812.64	51.89	4812.64	51.89	0.0005
24.WIF→EUR→USDT	324.72	10.12	324.72	10.12	324.72	10.12	0.0005
25.ADX→ETH→USDT	1.89	0.09	1.89	0.09	1.89	0.09	0.0005
26.1000SATS→USDC→USDT	59.92	2.37	59.92	2.37	59.92	2.37	0.0005
27.FET→BNB→USDT	127.73	5.44	127.73	5.44	127.73	5.44	0.0005
28.ETH→EURI→USDT	-47.95	-1.84	-47.95	-1.84	-47.95	-1.84	N/A
29.SOL→TUSD→USDT	37.62	2.85	37.62	2.85	37.62	2.85	0.0005
30.MEME→USDC→USDT	156.81	4.57	156.81	4.57	156.81	4.57	0.0005
31.TFUEL→BTC→USDT	326.29	15.11	326.29	15.11	326.29	15.11	0.0005
32.WIN→EUR→USDT	-100.00	-0.06	-100.00	-0.06	-100.00	-0.06	N/A
33.CTK→BNB→USDT	98.62	7.09	98.62	7.09	98.62	7.09	0.0005
34.ENA→BNB→USDT	280.30	10.40	280.30	10.40	280.30	10.40	0.0005
35.PYTH→USDC→USDT	24.41	0.63	24.41	0.63	24.41	0.63	0.0005
36.AI→BNB→USDT	170.41	10.35	170.41	10.35	170.41	10.35	0.0005
37.DASH→ETH→USDT	88.95	1.12	88.95	1.12	88.95	1.12	0.0005
38.HBAR→BNB→USDT	-100.00	-0.29	-100.00	-0.29	-100.00	-0.29	N/A
39.CHZ→BNB→USDT	-57.74	-1.26	-57.74	-1.26	-57.74	-1.26	N/A
40.ONT→USDC→USDT	139.12	7.41	139.12	7.41	139.12	7.41	0.0005
41.CATI→BNB→USDT	-99.97	-2.46	-99.97	-2.46	-99.97	-2.46	N/A
42.BAT→BTC→USDT	51.50	3.72	51.50	3.72	51.50	3.72	0.0005
43.GRT→EUR→USDT	161.51	10.46	161.51	10.46	161.51	10.46	0.0005
45.NEIRO→EUR→USDT	2017.48	157.52	2017.48	157.52	2017.48	157.52	0.0005
46.RARE→USDC→USDT	169.29	7.25	169.29	7.25	169.29	7.25	0.0005
47.ICP→EUR→USDT	142.45	7.25	142.45	7.25	142.45	7.25	0.0005
48.IMX→BTC→USDT	173.97	18.86	173.97	18.86	173.97	18.86	0.0005
49.RONIN→FDUSD→USDT	-100.00	-0.09	-100.00	-0.09	-100.00	-0.09	N/A
50.SUSHI→BTC→USDT	31.32	1.49	31.32	1.49	31.32	1.49	0.0005
51.DIA→BTC→USDT	48.16	4.25	48.16	4.25	48.16	4.25	0.0005
52.BNB→TUSD→USDT	-73.38	-2.26	-73.38	-2.26	-73.38	-2.26	N/A
53.YGG→USDC→USDT	142.83	7.52	142.83	7.52	142.83	7.52	0.0005
54.SFP→BTC→USDT	-62.45	-1.72	-62.45	-1.72	-62.45	-1.72	N/A
55.ZRX→BTC→USDT	33.09	2.94	33.09	2.94	33.09	2.94	0.0005
56.LTC→EUR→USDT	-100.00	-0.17	-100.00	-0.17	-100.00	-0.17	N/A
57.FLOKI→USDC→USDT	-100.00	-0.07	-100.00	-0.07	-100.00	-0.07	N/A
58.BTC→EURI→USDT	-100.00	-0.28	-100.00	-0.28	-100.00	-0.28	N/A
59.AAVE→ETH→USDT	-100.00	-0.33	-100.00	-0.33	-100.00	-0.33	N/A
60.MANA→ETH→USDT	-35.31	-2.21	-35.31	-2.21	-35.31	-2.21	N/A
62.AVAX→BNB→USDT	14.41	1.14	14.41	1.14	14.41	1.14	0.0005
63.EOS→ETH→USDT	7.10	1.05	7.10	1.05	7.10	1.05	0.0005
64.SXP→BTC→USDT	33.30	0.66	33.30	0.66	33.30	0.66	0.0005

Returns and Sharpe ratios across κ α_{edge} variations for all 64 strategies. Baseline values are indicated in the table.

“Optimal” is the value with highest Sharpe; N/A if all are negative.

Table B13: Parameter Sensitivity Analysis - $\kappa \alpha_{start}$ Variations

Strategy	2.5e-4 Ret(%)	2.5e-4 Sharpe	5e-4 Ret(%)	5e-4 Sharpe	1e-3 Ret(%)	1e-3 Sharpe	Optimal
01.XRP→TUSD→USDT	1.93	0.17	1.93	0.17	1.93	0.17	0.00025
02.INJ→BNB→USDT	64.81	4.84	64.81	4.84	64.81	4.84	0.00025
03.DOT→EUR→USDT	-0.95	-0.04	-0.95	-0.04	-0.95	-0.04	N/A
04.SUI→EUR→USDT	33.20	1.21	33.20	1.21	33.20	1.21	0.00025
05.LINK→EUR→USDT	28.40	1.49	28.40	1.49	28.40	1.49	0.00025
06.UNI→ETH→USDT	-89.49	-9.13	-89.49	-9.13	-89.49	-9.13	N/A
07.LISTA→BNB→USDT	215.10	10.64	215.10	10.64	215.10	10.64	0.00025
08.LRC→ETH→USDT	25.77	1.54	25.77	1.54	25.77	1.54	0.00025
09.IOTA→ETH→USDT	-56.52	-1.80	-56.52	-1.80	-56.52	-1.80	N/A
10.MANTA→USDC→USDT	113.76	6.73	113.76	6.73	113.76	6.73	0.00025
11.VET→EUR→USDT	144.19	7.83	144.19	7.83	144.19	7.83	0.00025
12.GALA→EUR→USDT	265.92	10.74	265.92	10.74	265.92	10.74	0.00025
13.MOVE→BNB→USDT	84.45	2.63	84.45	2.63	84.45	2.63	0.00025
14.POL→EUR→USDT	138.00	14.74	138.00	14.74	138.00	14.74	0.00025
15.PNUT→EUR→USDT	965.28	12.88	965.28	12.88	965.28	12.88	0.00025
16.AGLD→BTC→USDT	132.55	5.95	132.55	5.95	132.55	5.95	0.00025
17.REZ→USDC→USDT	259.99	9.25	259.99	9.25	259.99	9.25	0.00025
18.XLM→EUR→USDT	51.02	2.28	51.02	2.28	51.02	2.28	0.00025
19.UMA→BTC→USDT	69.57	2.45	69.57	2.45	69.57	2.45	0.00025
20.TRX→EUR→USDT	-100.00	-0.32	-100.00	-0.32	-100.00	-0.32	N/A
21.ARKM→BNB→USDT	295.47	12.47	295.47	12.47	295.47	12.47	0.00025
22.STEEM→ETH→USDT	42.83	3.34	42.83	3.34	42.83	3.34	0.00025
23.CELR→BTC→USDT	4812.64	51.89	4812.64	51.89	4812.64	51.89	0.00025
24.WIF→EUR→USDT	324.72	10.12	324.72	10.12	324.72	10.12	0.00025
25.ADX→ETH→USDT	1.89	0.09	1.89	0.09	1.89	0.09	0.00025
26.1000SATS→USDC→USDT	59.92	2.37	59.92	2.37	59.92	2.37	0.00025
27.FET→BNB→USDT	127.73	5.44	127.73	5.44	127.73	5.44	0.00025
28.ETH→EURI→USDT	-47.95	-1.84	-47.95	-1.84	-47.95	-1.84	N/A
29.SOL→TUSD→USDT	37.62	2.85	37.62	2.85	37.62	2.85	0.00025
30.MEME→USDC→USDT	156.81	4.57	156.81	4.57	156.81	4.57	0.00025
31.TFUEL→BTC→USDT	326.29	15.11	326.29	15.11	326.29	15.11	0.00025
32.WIN→EUR→USDT	-100.00	-0.06	-100.00	-0.06	-100.00	-0.06	N/A
33.CTK→BNB→USDT	98.62	7.09	98.62	7.09	98.62	7.09	0.00025
34.ENA→BNB→USDT	280.30	10.40	280.30	10.40	280.30	10.40	0.00025
35.PYTH→USDC→USDT	24.41	0.63	24.41	0.63	24.41	0.63	0.00025
36.AI→BNB→USDT	170.41	10.35	170.41	10.35	170.41	10.35	0.00025
37.DASH→ETH→USDT	88.95	1.12	88.95	1.12	88.95	1.12	0.00025
38.HBAR→BNB→USDT	-100.00	-0.29	-100.00	-0.29	-100.00	-0.29	N/A
39.CHZ→BNB→USDT	-57.74	-1.26	-57.74	-1.26	-57.74	-1.26	N/A
40.ONT→USDC→USDT	139.12	7.41	139.12	7.41	139.12	7.41	0.00025
41.CATI→BNB→USDT	-99.97	-2.46	-99.97	-2.46	-99.97	-2.46	N/A
42.BAT→BTC→USDT	51.50	3.72	51.50	3.72	51.50	3.72	0.00025
43.GRT→EUR→USDT	161.51	10.46	161.51	10.46	161.51	10.46	0.00025
45.NEIRO→EUR→USDT	2017.48	157.52	2017.48	157.52	2017.48	157.52	0.00025
46.RARE→USDC→USDT	169.29	7.25	169.29	7.25	169.29	7.25	0.00025
47.ICP→EUR→USDT	142.45	7.25	142.45	7.25	142.45	7.25	0.00025
48.IMX→BTC→USDT	173.97	18.86	173.97	18.86	173.97	18.86	0.00025
49.RONIN→FDUSD→USDT	-100.00	-0.09	-100.00	-0.09	-100.00	-0.09	N/A
50.SUSHI→BTC→USDT	31.32	1.49	31.32	1.49	31.32	1.49	0.00025
51.DIA→BTC→USDT	48.16	4.25	48.16	4.25	48.16	4.25	0.00025
52.BNB→TUSD→USDT	-73.38	-2.26	-73.38	-2.26	-73.38	-2.26	N/A
53.YGG→USDC→USDT	142.83	7.52	142.83	7.52	142.83	7.52	0.00025
54.SFP→BTC→USDT	-62.45	-1.72	-62.45	-1.72	-62.45	-1.72	N/A
55.ZRX→BTC→USDT	33.09	2.94	33.09	2.94	33.09	2.94	0.00025
56.LTC→EUR→USDT	-100.00	-0.17	-100.00	-0.17	-100.00	-0.17	N/A
57.FLOKI→USDC→USDT	-100.00	-0.07	-100.00	-0.07	-100.00	-0.07	N/A
58.BTC→EURI→USDT	-100.00	-0.28	-100.00	-0.28	-100.00	-0.28	N/A
59.AAVE→ETH→USDT	-100.00	-0.33	-100.00	-0.33	-100.00	-0.33	N/A
60.MANA→ETH→USDT	-35.31	-2.21	-35.31	-2.21	-35.31	-2.21	N/A
62.AVAX→BNB→USDT	14.41	1.14	14.41	1.14	14.41	1.14	0.00025
63.EOS→ETH→USDT	7.10	1.05	7.10	1.05	7.10	1.05	0.00025
64.SXP→BTC→USDT	33.30	0.66	33.30	0.66	33.30	0.66	0.00025

Returns and Sharpe ratios across $\kappa \alpha_{start}$ variations for all 64 strategies. Baseline values are indicated in the table.

“Optimal” is the value with highest Sharpe; N/A if all are negative.

Table B14: Parameter Sensitivity Analysis - Slippage Weight Variations

Strategy	0.5 Ret(%)	0.5 Sharpe	0.6 Ret(%)	0.6 Sharpe	0.7 Ret(%)	0.7 Sharpe	Optimal
01.XRP→TUSD→USDT	1.39	0.12	1.93	0.17	2.03	0.18	0.7
02.INJ→BNB→USDT	64.04	4.78	64.81	4.84	66.69	5.03	0.7
03.DOT→EUR→USDT	-1.47	-0.06	-0.95	-0.04	0.12	0.01	0.7
04.SUI→EUR→USDT	33.04	1.20	33.20	1.21	34.19	1.25	0.7
05.LINK→EUR→USDT	24.90	1.31	28.40	1.49	29.30	1.53	0.7
06.UNI→ETH→USDT	-90.15	-7.74	-89.49	-9.13	-87.27	-18.98	N/A
07.LISTA→BNB→USDT	214.97	10.63	215.10	10.64	218.74	10.89	0.7
08.LRC→ETH→USDT	25.82	1.54	25.77	1.54	29.34	1.80	0.7
09.IOTA→ETH→USDT	-56.48	-1.80	-56.52	-1.80	-53.20	-1.78	N/A
10.MANTA→USDC→USDT	113.81	6.74	113.76	6.73	113.98	6.75	0.7
11.VET→EUR→USDT	144.30	7.83	144.19	7.83	145.17	7.90	0.7
12.GALA→EUR→USDT	266.03	10.74	265.92	10.74	266.90	10.76	0.7
13.MOVE→BNB→USDT	84.33	2.63	84.45	2.63	87.54	2.75	0.7
14.POL→EUR→USDT	138.08	14.74	138.00	14.74	138.89	14.74	0.5
15.PNUT→EUR→USDT	965.41	12.89	965.28	12.88	967.72	12.91	0.7
16.AGLD→BTC→USDT	130.95	5.88	132.55	5.95	134.52	6.02	0.7
17.REZ→USDC→USDT	260.11	9.25	259.99	9.25	260.50	9.26	0.7
18.XLM→EUR→USDT	51.13	2.29	51.02	2.28	52.10	2.33	0.7
19.UMA→BTC→USDT	68.99	2.42	69.57	2.45	69.90	2.47	0.7
20.TRX→EUR→USDT	-100.00	-0.31	-100.00	-0.32	-100.00	-0.04	N/A
21.ARKM→BNB→USDT	295.37	12.46	295.47	12.47	298.64	12.56	0.7
22.STEEM→ETH→USDT	42.83	3.35	42.83	3.34	44.83	3.54	0.7
23.CELR→BTC→USDT	4811.64	51.89	4812.64	51.89	4864.99	52.43	0.7
24.WIF→EUR→USDT	324.61	10.12	324.72	10.12	325.55	10.13	0.7
25.ADX→ETH→USDT	1.92	0.09	1.89	0.09	3.80	0.18	0.7
26.1000SATS→USDC→USDT	60.08	2.37	59.92	2.37	60.25	2.38	0.7
27.FET→BNB→USDT	127.63	5.43	127.73	5.44	130.14	5.61	0.7
28.ETH→EURI→USDT	-47.96	-1.84	-47.95	-1.84	-47.90	-1.84	N/A
29.SOL→TUSD→USDT	37.23	2.81	37.62	2.85	37.64	2.85	0.7
30.MEME→USDC→USDT	156.94	4.57	156.81	4.57	157.34	4.58	0.7
31.TFUEL→BTC→USDT	325.84	15.11	326.29	15.11	329.58	15.24	0.7
32.WIN→EUR→USDT	-100.00	-0.04	-100.00	-0.06	-100.00	-0.16	N/A
33.CTK→BNB→USDT	98.52	7.08	98.62	7.09	100.79	7.27	0.7
34.ENA→BNB→USDT	280.22	10.40	280.30	10.40	282.91	10.51	0.7
35.PYTH→USDC→USDT	24.50	0.63	24.41	0.63	24.83	0.64	0.7
36.AI→BNB→USDT	170.36	10.35	170.41	10.35	173.35	10.48	0.7
37.DASH→ETH→USDT	88.16	1.11	88.95	1.12	91.32	1.18	0.7
38.HBAR→BNB→USDT	-100.00	-0.29	-100.00	-0.29	-100.00	-0.33	N/A
39.CHZ→BNB→USDT	-57.72	-1.26	-57.74	-1.26	-54.66	-1.24	N/A
40.ONT→USDC→USDT	139.22	7.42	139.12	7.41	139.57	7.43	0.7
41.CATI→BNB→USDT	-99.97	-2.45	-99.97	-2.46	-99.97	-2.46	N/A
42.BAT→BTC→USDT	50.45	3.62	51.50	3.72	53.23	3.91	0.7
43.GRT→EUR→USDT	161.61	10.46	161.51	10.46	162.36	10.49	0.7
44.BOME→EUR→USDT	704.40	23.76	704.23	23.75	705.78	23.83	0.7
45.NEIRO→EUR→USDT	2018.89	157.66	2017.48	157.52	2032.26	159.29	0.7
46.RARE→USDC→USDT	169.37	7.25	169.29	7.25	169.65	7.26	0.7
47.ICP→EUR→USDT	141.43	7.20	142.45	7.25	143.20	7.31	0.7
48.IMX→BTC→USDT	172.47	18.80	173.97	18.86	175.76	18.94	0.7
49.RONIN→FDUSD→USDT	-100.00	-0.09	-100.00	-0.09	-100.00	-0.09	N/A
50.SUSHI→BTC→USDT	29.27	1.38	31.32	1.49	33.32	1.59	0.7
51.DIA→BTC→USDT	46.78	4.11	48.16	4.25	49.70	4.44	0.7
52.BNB→TUSD→USDT	-73.54	-2.25	-73.38	-2.26	-73.38	-2.26	N/A
53.YGG→USDC→USDT	142.92	7.53	142.83	7.52	143.28	7.54	0.7
54.SFP→BTC→USDT	-63.23	-1.72	-62.45	-1.72	-59.80	-1.72	N/A
55.ZRX→BTC→USDT	32.10	2.86	33.09	2.94	34.79	3.06	0.7
56.LTC→EUR→USDT	-100.00	-0.21	-100.00	-0.17	-100.00	-0.22	N/A
57.FLOKI→USDC→USDT	-100.00	-0.17	-100.00	-0.07	-100.00	-0.27	N/A
58.BTC→EURI→USDT	-100.00	-0.28	-100.00	-0.28	-100.00	-0.27	N/A
59.AAVE→ETH→USDT	-100.00	-0.39	-100.00	-0.33	-100.00	-0.27	N/A
60.MANA→ETH→USDT	-35.32	-2.21	-35.31	-2.21	-32.24	-2.07	N/A
61.DOGE→TUSD→USDT	88.22	9.45	88.51	9.48	88.66	9.50	0.7
62.AVAX→BNB→USDT	13.47	1.05	14.41	1.14	15.21	1.21	0.7
63.EOS→ETH→USDT	6.93	1.03	7.10	1.05	9.53	1.42	0.7
64.SXP→BTC→USDT	32.38	0.64	33.30	0.66	35.58	0.72	0.7

Returns and Sharpe ratios across slippage weight variations for all 64 strategies. Baseline values are indicated in the table. "Optimal" is the value with highest Sharpe; N/A if all are negative.

Table B15: Transaction Cost Sensitivity Analysis - Part 1 (Cost Levels 0.01%-0.10%)

Strategy	0.01%	0.02%	0.03%	0.04%	0.05%	0.06%	0.07%	0.08%	0.09%	0.10%	BE	Res
01.XRP→TUSD→USDT	144.32%	108.14%	72.29%	36.86%	1.93%	-32.31%	-65.53%	-96.60%	-100.00%	-100.00%	≈0.06%	Low
02.INJ→BNB→USDT	234.83%	191.80%	149.08%	106.72%	64.81%	23.46%	-17.13%	-56.58%	-93.71%	-100.00%	≈0.07%	Medium
03.DOT→EUR→USDT	351.72%	261.98%	172.94%	85.03%	-0.95%	-82.58%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.05%	None
04.SUI→EUR→USDT	369.88%	283.90%	198.89%	115.15%	33.20%	-45.80%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
05.LINK→EUR→USDT	334.39%	256.30%	179.06%	102.95%	28.40%	-43.65%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
06.UNI→ETH→USDT	129.71%	72.66%	16.59%	-37.96%	-89.49%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
07.LISTA→BNB→USDT	457.82%	396.18%	335.12%	274.72%	215.10%	156.41%	98.86%	42.76%	-11.34%	-62.35%	≈0.09%	Medium
08.LRC→ETH→USDT	234.62%	181.50%	128.89%	76.92%	25.77%	-24.22%	-72.23%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
09.IOTA→ETH→USDT	147.46%	95.01%	43.29%	-7.41%	-56.52%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
10.MANTA→USDC→USDT	256.22%	220.33%	184.60%	149.07%	113.76%	78.73%	44.03%	9.76%	-23.94%	-56.80%	≈0.09%	Medium
11.VET→EUR→USDT	392.32%	329.67%	267.37%	205.50%	144.19%	83.61%	24.04%	-33.98%	-88.81%	-100.00%	≈0.08%	Medium
12.GALA→EUR→USDT	454.43%	407.03%	359.79%	312.74%	265.92%	219.36%	173.10%	127.22%	81.79%	36.95%	≈0.11%	Medium
13.MOVE→BNB→USDT	292.80%	240.02%	187.65%	135.75%	84.45%	33.91%	-15.56%	-63.34%	-100.00%	-100.00%	≈0.07%	Medium
14.POL→EUR→USDT	340.66%	289.60%	238.76%	188.21%	138.00%	88.25%	39.11%	-9.15%	-56.02%	-99.34%	≈0.08%	Medium
15.PNUT→EUR→USDT	1258.63%	1185.09%	1111.68%	1038.41%	965.28%	892.34%	819.58%	747.04%	674.75%	602.73%	≈0.19%	High
16.AGLD→BTC→USDT	317.24%	270.74%	224.42%	178.34%	132.55%	87.14%	42.23%	-1.98%	-45.13%	-86.26%	≈0.08%	Medium
17.REZ→USDC→USDT	566.90%	489.49%	412.48%	335.94%	259.99%	184.79%	110.56%	37.71%	-33.00%	-98.43%	≈0.09%	Medium
18.XLM→EUR→USDT	345.01%	270.35%	196.30%	123.08%	51.02%	-19.22%	-85.70%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
19.UMA→BTC→USDT	305.16%	245.41%	186.14%	127.47%	69.57%	12.72%	-42.52%	-94.24%	-100.00%	-100.00%	≈0.07%	Medium
20.TRX→EUR→USDT	57.78%	0.52%	-54.85%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
21.ARKM→BNB→USDT	522.62%	465.67%	408.80%	352.05%	295.47%	239.11%	183.05%	127.39%	72.30%	18.01%	≈0.11%	Medium
22.STEEM→ETH→USDT	206.42%	165.02%	123.90%	83.14%	42.83%	3.09%	-35.81%	-73.36%	-100.00%	-100.00%	≈0.07%	Medium
23.CELR→BTC→USDT	10399.86%	8712.32%	7228.94%	5934.13%	4812.64%	3849.67%	3030.80%	2342.09%	1770.10%	1301.84%	≈0.17%	High
24.WIF→EUR→USDT	498.35%	454.80%	411.34%	367.97%	324.72%	281.61%	238.68%	195.94%	153.46%	111.28%	≈0.13%	Medium
25.ADX→ETH→USDT	164.93%	123.51%	82.45%	41.87%	1.89%	-37.22%	-74.91%	-100.00%	-100.00%	-100.00%	≈0.06%	Low
26.1000SATS→USDC→USDT	239.64%	194.10%	148.91%	104.15%	59.92%	16.36%	-26.28%	-67.46%	-100.00%	-100.00%	≈0.07%	Medium
27.FET→BNB→USDT	324.78%	274.97%	225.49%	176.38%	127.73%	79.64%	32.25%	-14.19%	-59.15%	-100.00%	≈0.08%	Medium
28.ETH→EURI→USDT	118.21%	75.75%	33.77%	-7.58%	-47.95%	-86.45%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
29.SOL→TUSD→USDT	202.66%	160.83%	119.33%	78.23%	37.62%	-2.33%	-41.38%	-78.87%	-100.00%	-100.00%	≈0.06%	Medium
30.MEME→USDC→USDT	495.02%	409.17%	324.07%	239.88%	156.81%	75.25%	-4.13%	-79.36%	-100.00%	-100.00%	≈0.07%	Medium
31.TFUEL→BTC→USDT	614.33%	542.02%	469.86%	397.93%	326.29%	255.05%	184.34%	114.37%	45.46%	-21.78%	≈0.10%	Medium
32.WIN→EUR→USDT	-60.52%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.01%	None
33.CTK→BNB→USDT	304.62%	252.51%	200.76%	149.42%	98.62%	48.48%	-0.74%	-48.58%	-93.52%	-100.00%	≈0.07%	Medium
34.ENA→BNB→USDT	465.47%	418.92%	372.53%	326.31%	280.30%	234.52%	189.02%	143.85%	99.08%	54.82%	≈0.12%	Medium
35.PYTH→USDC→USDT	338.71%	258.49%	179.14%	100.95%	24.41%	-49.41%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
36.AI→BNB→USDT	364.61%	315.63%	266.90%	218.48%	170.41%	122.77%	75.66%	29.22%	-16.30%	-60.38%	≈0.09%	Medium
37.DASH→ETH→USDT	311.87%	255.35%	199.29%	143.78%	88.95%	34.99%	-17.73%	-68.44%	-100.00%	-100.00%	≈0.07%	Medium
38.HBAR→BNB→USDT	203.98%	107.83%	13.91%	-75.42%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
39.CHZ→BNB→USDT	141.59%	90.30%	39.76%	-9.78%	-57.74%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
40.ONT→USDC→USDT	350.20%	296.86%	243.86%	191.25%	139.12%	87.58%	36.80%	-12.94%	-61.01%	-100.00%	≈0.08%	Medium
41.CATI→BNB→USDT	-99.61%	-99.79%	-99.89%	-99.94%	-99.97%	-99.99%	-99.99%	-100.00%	-100.00%	-100.00%	≈0.01%	None
42.BAT→BTC→USDT	251.29%	200.63%	150.37%	100.62%	51.50%	3.23%	-43.77%	-88.31%	-100.00%	-100.00%	≈0.07%	Medium
43.GRT→EUR→USDT	334.80%	291.07%	247.59%	204.39%	161.51%	119.01%	76.95%	35.45%	-5.33%	-45.09%	≈0.09%	Medium
44.BOME→EUR→USDT	1018.90%	938.42%	859.10%	781.01%	704.23%	628.84%	554.94%	482.65%	412.11%	343.47%	≈0.16%	High
45.NEIRO→EUR→USDT	5563.34%	4386.73%	3425.18%	2645.19%	2017.48%	1516.62%	1120.65%	810.68%	570.65%	386.94%	≈0.15%	Medium
46.RARE→USDC→USDT	368.51%	318.28%	268.29%	218.61%	169.29%	120.40%	72.06%	24.43%	-22.20%	-67.20%	≈0.09%	Medium
47.ICP→EUR→USDT	310.69%	268.20%	225.97%	184.04%	142.45%	101.26%	60.54%	20.41%	-18.94%	-57.15%	≈0.09%	Medium
48.IMX→BTC→USDT	347.76%	304.04%	260.48%	217.11%	173.97%	131.12%	88.62%	46.57%	5.13%	-35.44%	≈0.10%	Medium
49.RONIN→FDUSD→USDT	107.16%	50.09%	-5.79%	-59.75%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
50.SUSHI→BTC→USDT	255.28%	198.40%	142.01%	86.24%	31.32%	-22.36%	-73.87%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
51.DIA→BTC→USDT	207.84%	167.42%	127.28%	87.50%	48.16%	9.37%	-28.66%	-65.51%	-99.51%	-100.00%	≈0.07%	Medium
52.BNB→TUSD→USDT	106.72%	60.21%	14.42%	-30.33%	-73.38%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
53.YGG→USDC→USDT	373.37%	315.01%	257.09%	199.66%	142.83%	86.74%	31.60%	-22.20%	-73.72%	-100.00%	≈0.08%	Medium
54.SFP→BTC→USDT	148.34%	94.03%	40.52%	-11.87%	-62.45%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
55.ZRX→BTC→USDT	205.24%	161.14%	117.66%	74.92%	33.09%	-7.57%	-46.59%	-82.86%	-100.00%	-100.00%	≈0.06%	Medium
56.LTC→EUR→USDT	206.05%	99.14%	-5.02%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
57.FLOKI→USDC→USDT	183.97%	62.34%	-53.88%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
58.BTC→EURI→USDT	32.00%	-14.49%	-59.53%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.02%	None
59.AAVE→ETH→USDT	87.49%	29.19%	-27.70%	-81.90%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
60.MANA→ETH→USDT	174.23%	120.54%	67.53%	15.44%	-35.31%	-83.61%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.05%	None
61.DOGE→TUSD→USDT	272.83%	225.36%	178.73%	133.06%	88.51%	45.30%	3.75%	-35.63%	-71.79%	-100.00%	≈0.08%	Medium
62.AVAX→BNB→USDT	198.48%	151.69%	105.33%	59.51%	14.41%	-29.70%	-72.18%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
63.EOS→ETH→USDT	289.20%	197.89%	121.06%	57.79%	7.10%	-32.09%	-60.90%	-80.60%	-92.56%	-98.38%	≈0.06%	Medium
64.SXP→BTC→USDT	241.01%	188.31%	136.04%	84.31%	33.30%	-16.67%	-64.92%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium

Table B16: Transaction Cost Sensitivity Analysis - Part 2 (Cost Levels 0.11%–0.20%)

Strategy	0.11%	0.12%	0.13%	0.14%	0.15%	0.16%	0.17%	0.18%	0.19%	0.20%	BE	Res
01.XRP→TUSD→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Low
02.INJ→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
03.DOT→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.05%	None
04.SUI→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
05.LINK→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
06.UNI→ETH→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
07.LISTA→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.09%	Medium
08.LRC→ETH→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
09.IOTA→ETH→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
10.MANTA→USDC→USDT	-88.14%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.09%	Medium
11.VET→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.08%	Medium
12.GALA→EUR→USDT	-7.11%	-49.98%	-90.54%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.11%	Medium
13.MOVE→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
14.POL→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.08%	Medium
15.PNUT→EUR→USDT	531.03%	459.69%	388.77%	318.34%	248.49%	179.37%	111.14%	44.12%	-21.12%	-82.98%	≈0.19%	High
16.AGLD→BTC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.08%	Medium
17.REZ→USDC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.09%	Medium
18.XLM→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
19.UMA→BTC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
20.TRX→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
21.ARKM→BNB→USDT	-35.01%	-85.47%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.11%	Medium
22.STEEM→ETH→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
23.CELR→BTC→USDT	924.91%	627.44%	398.17%	226.48%	102.43%	16.80%	-38.85%	-72.13%	-89.75%	-97.41%	≈0.17%	High
24.WIF→EUR→USDT	69.49%	28.19%	-12.42%	-52.00%	-89.57%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.13%	Medium
25.ADX→ETH→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Low
26.1000SATS→USDC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
27.FET→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.08%	Medium
28.ETH→EURI→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
29.SOL→TUSD→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
30.MEME→USDC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
31.TFUEL→BTC→USDT	-85.52%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.10%	Medium
32.WIN→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.01%	None
33.CTK→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
34.ENA→BNB→USDT	11.23%	-31.42%	-72.52%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.12%	Medium
35.PYTH→USDC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
36.AI→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.09%	Medium
37.DASH→ETH→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
38.HBAR→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
39.CHZ→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
40.ONT→USDC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.08%	Medium
41.CATI→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.01%	None
42.BAT→BTC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
43.GRT→EUR→USDT	-83.07%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.09%	Medium
44.BOME→EUR→USDT	276.91%	212.67%	151.05%	92.44%	37.37%	-13.34%	-58.24%	-93.73%	-100.00%	-100.00%	≈0.16%	High
45.NEIRO→EUR→USDT	248.11%	144.67%	68.77%	14.01%	-24.74%	-51.60%	-69.78%	-81.75%	-89.39%	-94.11%	≈0.15%	Medium
46.RARE→USDC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.09%	Medium
47.ICP→EUR→USDT	-93.14%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.09%	Medium
48.IMX→BTC→USDT	-74.54%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.10%	Medium
49.RONIN→FDUSD→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
50.SUSHI→BTC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
51.DIA→BTC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.07%	Medium
52.BNB→TUSD→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
53.YGG→USDC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.08%	Medium
54.SFP→BTC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.04%	None
55.ZRX→BTC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
56.LTC→EUR→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
57.FLOKI→USDC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
58.BTC→EURI→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.02%	None
59.AAVE→ETH→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.03%	None
60.MANA→ETH→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.05%	None
61.DOGE→TUSD→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.08%	Medium
62.AVAX→BNB→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
63.EOS→ETH→USDT	-99.97%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium
64.SXP→BTC→USDT	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	-100.00%	≈0.06%	Medium